DiPaSSCD: Difference-Paired Self-Supervised Learning With Object-Based Local Augmentation for Remote Sensing Image Change Detection

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***Abstract*—Change detection (CD) from remote sensing imagery is an essential task for monitoring various Earth dynamics. Recently, self-supervised learning (SSL) has emerged as a promising paradigm to alleviate the reliance on large-scale annotated datasets. Previous SSL methods for CD suffer from spatial misalignment caused by multiview augmentation, inherent tradeoffs among SSL paradigms, and the lack of explicit change semantics in representation. To overcome these challenges, we propose difference-paired self-supervised learning for CD (DiPaSSCD), a novel pretraining framework that explicitly integrates change-oriented signals into SSL. The method consists of three key components: 1) object-based local (OBL) augmentation, which simulates realistic local variations while preserving spatial alignment; 2) mask-guided SSL integration, which jointly optimizes global contrastive and dense prediction objectives to mitigate the limitations of each paradigm; and 3) difference-paired contrastive learning (DiPaCL), which directly learns embeddings from paired inputs containing explicit differences, yielding inherently change-related feature representations. Extensive experiments on LEVIR-CD, S2Looking, and DSIFN datasets under both full and label-limited conditions demonstrate that DiPaSSCD consistently outperforms supervised and SSL-based baselines. Qualitative analyses further confirm its ability to suppress pseudo-changes induced by illumination variations while capturing subtle, fine-scale land-cover changes. These results highlight the potential of DiPaSSCD as a versatile and architecture-agnostic pretraining paradigm for robust CD across diverse scenarios.**

***Index Terms*—Change detection (CD), contrastive learning (CL), mask-guided learning, object-based local (OBL) augmentation, remote sensing, self-supervised learning (SSL).**

I. Introduction

CHANGE detection (CD) using remote sensing imagery has become a fundamental task in Earth observation, enabling the monitoring of land-use dynamics, urban expansion, deforestation, and disaster impacts [1]. With the growing availability of high-resolution satellite data, deep learning (DL)-based approaches have shown remarkable performance in CD tasks. A wide range of supervised learning models have achieved state-of-the-art (SOTA) results by leveraging sufficiently annotated change maps [2], [3], [4]. However, such approaches fundamentally rely on large-scale labeled data, which are costly and time-consuming to obtain, particularly for bi-temporal or multitemporal imagery.

Self-supervised learning (SSL) has gained increasing attention in recent years as a promising alternative to these limitations. SSL methods can be broadly categorized into contrastive learning or generative approaches [5]. Contrastive learning (CL) aims to learn robust representations by embedding semantically similar (positive) and dissimilar (negative) pairs, typically generated via multiview data augmentation, into a feature space. This paradigm has demonstrated superior representation performance, owing to its ability to learn discriminative features without human annotations. Several successful frameworks such as Momentum Contrast (MoCo) [6], [7], SimCLR [8], SimSiam [9], and BYOL [10] have been proposed based on this paradigm. In contrast, early generative approaches [11], [12], [13], such as solving jigsaw puzzles or predicting image rotations, were initially less effective than CL. However, with the emergence of Transformer architectures and their successful adoption in the natural language processing (NLP) domain, self-prediction approaches [14], [15], [16], such as masked image modeling (MIM), have recently achieved remarkable performance in the computer vision field as well. These methods offer advantages in dense representation learning by reconstructing missing or corrupted parts of input images, making them particularly effective for pixel-level tasks.

In recent years, SSL has been increasingly applied to CD tasks, not only through simple transfer strategies using general-purpose SSL models but also via the design of change-specific pretext tasks. By leveraging supervision signals intrinsic to the data, SSL facilitates robust and generalizable change representation learning under limited or noisy annotations, which has been recognized as a promising direction in recent survey studies [17]. Many studies have explored how to design self-supervision signals that encourage the model to extract features relevant to land-cover changes while remaining invariant to irrelevant noise. One representative line of work is based on CL [Fig. 1(a)] [18], [19], [20], [21], [22], [23], [24], [25], [26]. Inspired by original CL frameworks, these methods define self-supervision through similarity objectives by feeding artificially augmented image pairs, such as those generated

Received 15 October 2025; revised 13 January 2026; accepted 29 January 2026. Date of publication 3 February 2026; date of current version 20 February 2026. This work was supported by the National Research Foundation of Korea (NRF) through Korean Government, Ministry of Science and ICT (MSIT) under Grant NRF-2023R1A2C2005548 and Grant RS-2025-16071712. *(Corresponding author: Yongil Kim.)*

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Digital Object Identifier 10.1109/TGRS.2026.3660265

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with radiometric perturbations and geometric transformations. The idea is that the model learns to identify semantically consistent patterns from artificially augmented images while ignoring pseudo-changes such as variations in illumination conditions or sensor geometry. Nevertheless, this approach has a notable limitation in that it usually produces global features with coarse spatial resolution, thereby limiting its suitability for dense prediction tasks such as CD. Moreover, the multiview augmentation scheme commonly used in CL introduces spatial misalignment between views, making it ill-suited for CD tasks that aim to identify local changes from geometrically registered bi-temporal images.

Alongside CL methods, generative approaches based on self-prediction or reconstruction strategies [Fig. 1(b)] have also been introduced into the CD task [27], [28], [29], [30]. These approaches typically involve randomly masking patches of the input image and training the model to reconstruct the missing content, thereby enhancing its contextual and generative understanding. When applied to CD, such methods promote the learning of robust representations that are less sensitive to irrelevant changes or background noise. Importantly, in contrast to CL frameworks, generative approaches allow the model to learn pixel-level representations by reconstructing the masked image back to the original input scale. However, the reconstruction-guided features do not encode change-related semantics between bi-temporal images, but instead reflect inherent appearance information derived from single-image representations. Consequently, detecting changes inevitably relies on a double-stream strategy based on a Siamese architecture or on additional change-specific decoders.

Another line of research has sought to design more change-related supervision by using pseudo-labels that directly reflect interimage differences [Fig. 1(c)] [31], [32], [33], [34], [35], [36], [37], [38]. These methods involve generating high-confidence initial change samples using techniques such as image differencing or unsupervised clustering and then training a model to learn feature representations related to the changes from these samples. Recently, several studies have extended this idea by incorporating pseudo-labels into mask-based reconstruction approaches [39], [40]. These methods introduce synthetic change masks that simulate changed areas, which are treated as pseudo-labels for self-supervision. Unlike the earlier two approaches, pseudo-label approaches take paired bi-temporal images as input, not only enabling the model to extract change-related features directly but also allowing structural interaction between the two inputs within a joint feature learning stream. Nevertheless, these approaches face several limitations, particularly related to the quality and reliability of the initial pseudo-labels or masks. Inaccurate or low-level pseudo-labels can introduce misleading supervision, potentially biasing the model toward incorrect change signals. Additionally, these approaches often emphasize the inherent local consistency, which may result in an inferior global discriminative power compared to the CL framework.

Despite these previous attempts, existing SSL-based CD approaches still exhibit three research limitations. First, the multiview augmentation strategy commonly adopted in CL is inherently inconsistent with the CD objectives, where local changes are detected from the spatially aligned bi-temporal images. Second, while contrastive and generative paradigms suffer from inherent tradeoffs. CL predominantly learns global and discriminative representations but lacks dense pixel-level semantics, whereas the latter emphasizes generative reconstruction without explicit discriminative capability. Third, most SSL CD frameworks are not explicitly designed to capture change-related semantics. Instead, they primarily focus on modeling invariant feature representations from single images and subsequently rely on Siamese-based feature differencing to identify changed areas.

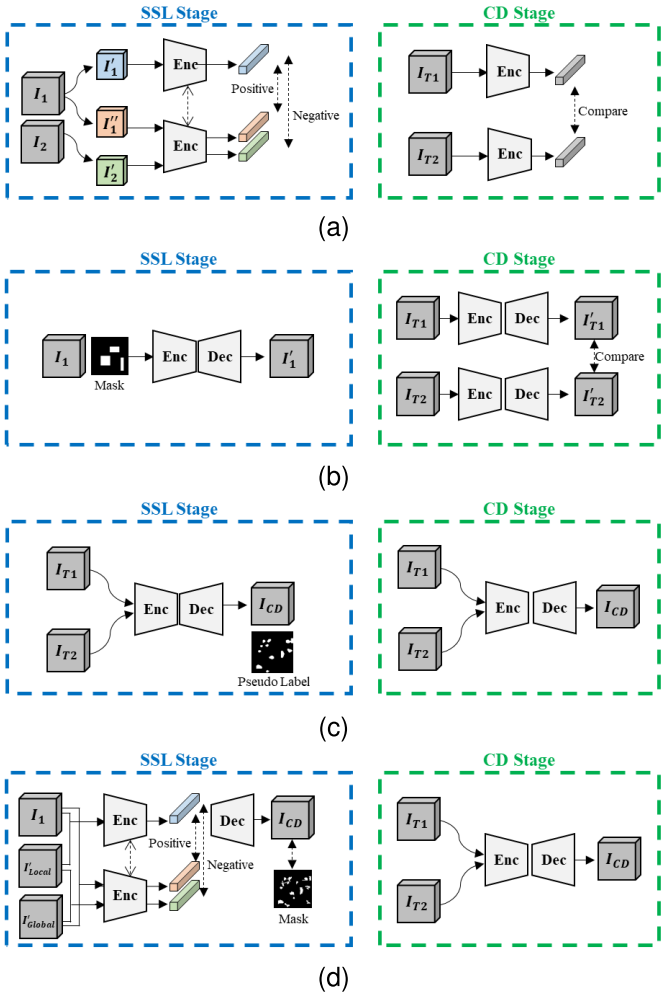


Fig. 1. Comparison of SSL paradigms for CD, including both the pretraining (SSL stage) and downstream task (CD stage). (a) CL-based approach, (b) generative reconstruction-based approach, (c) pseudo-label-based approach, and (d) proposed DiPaSSCD approach.

To address these limitations, this study proposes difference-paired self-supervised learning for change detection (DiPaSSCD) [Fig. 1(d)], a novel pretraining framework that jointly integrates CL and dense prediction objectives under a difference-oriented design. The key contributions of this work can be summarized as follows.

1) *Object-Based Local (OBL) Augmentation:* We introduce a change-related augmentation strategy on the object-based image analysis (OBIA) paradigm and a local

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augmentation scheme to simulate diverse local variations from a single-temporal image. This design addresses the misalignment of conventional multiview augmentations and enables the model to better distinguish land-cover changes from pseudo-changes caused by appearance variations.

2) *Mask-Guided SSL Integration:* We integrate CL and mask-guided dense prediction, where CL enforces global discriminative representations while the mask-guided branch provides dense contextual cues. This joint optimization mitigates the inherent tradeoffs of the previous CL approach and facilitates learning of both dense and discriminative feature representations.
3) *Change-Related Feature Representation by Difference-Paired Contrastive Learning (DiPaCL):* Instead of relying on feature comparison within Siamese backbones, DiPaCL directly learns embeddings from difference-paired inputs. Such inherently change-related representations not only enhance sensitivity to subtle variations but also provide a versatile pretraining paradigm that is agnostic to specific architectural constraints and readily adaptable to diverse CD backbones.

The remainder of this article is organized as follows. Section II reviews the related works on remote sensing CD and SSL for CD. Section III introduces the proposed DiPaSSCD framework, including OBL augmentation and the DiPaCL approach. Section IV presents the experimental settings, datasets, comparison methods, and quantitative as well as qualitative results on benchmark datasets. Section V discusses the adaptability, change-related feature interpretability, and ablation analyses of the proposed approach. Finally, Section VI concludes this article and outlines future research directions.

II. Related Works

A. Remote Sensing CD

Early CD methods are primarily based on pixel-wise algebraic operations between bi-temporal images, such as image differencing, followed by thresholding to identify changed regions [1]. A representative example is change vector analysis (CVA), originally proposed in [41], which characterizes changes by mapping both the magnitude and direction of spectral change vectors. Despite their conceptual simplicity and low computational cost, algebra-based methods are highly sensitive to noise, often resulting in salt-and-pepper artifacts, and generally struggle to discriminate complex or nonlinear change patterns. Also, transformation-based approaches [42], [43], [44], such as principal component analysis (PCA), have been employed to reduce band redundancy and enhance change-related information by projecting the original data into a transformed feature space [1]. While the transformation-based methods are advantageous for compressing high-dimensional data and can capture certain nonlinear change characteristics, the resulting transformed features are often scene-dependent, which limits their generalizability across different images or acquisition conditions. Overall, these conventional CD approaches predominantly rely on handcrafted features and manually designed decision criteria, which constrains their ability to model complex semantic changes [45]. Although effective for coarse-resolution imagery, their performance degrades significantly when applied to very high-resolution (VHR) remote sensing data, where fine spatial details, complex spatial patterns, and high intraclass spectral variability substantially compromise their robustness.

With the rapid progress of DL in computer vision, DL-based CD approaches have emerged as powerful alternatives, demonstrating superior capability in learning high-level feature representations and achieving substantially improved detection accuracy. Among them, Siamese architectures [46], [47], [48], [49], [50] have been widely adopted, in which two parallel branches extract features from temporally paired images, and a subsequent feature differencing or fusion stage identifies changes. Representatively, Daudt et al. [49] proposed three fully convolutional neural networks—FC-EF, FC-Siam-conc, and FC-Siam-diff—demonstrating the effectiveness of Siamese architecture for bi-temporal CD. These architectures extended conventional convolutional networks to a fully convolutional paradigm with skip connections and proposed two Siamese variants that integrate bi-temporal features through early, mid, and late fusion schemes. This design significantly improved both detection accuracy and computational efficiency compared to previous methods. Building upon this foundation, Fang et al. [50] introduced a more advanced Siamese architecture that incorporates nested UNet modules, enabling hierarchical feature extraction and refinement. This design effectively captured multiscale context and local structural details, leading to improved accuracy, especially in densely built environments.

Recent studies have further refined the Siamese paradigm by introducing explicit feature interactions across the two temporal branches at the encoder stage [51], [52], [53], [54], [55]. Especially, Fang et al. [54] proposed MetaChanger, a general CD framework that emphasizes feature interaction between bi-temporal features. It introduced two effective strategies—aggregation distribution and feature exchange—to enhance interaction during feature extraction. These designs collectively improved temporal feature alignment and yielded competitive performance across multiple benchmarks. Additionally, Codegoni et al. [55] proposed a lightweight model, TinyCD, by leveraging low-level features and introducing an efficient spatiotemporal mixing strategy. TinyCD significantly reduced model size and complexity while maintaining competitive accuracy on public CD benchmarks.

More recently, Transformer architectures have been introduced to CD tasks, leveraging their global receptive field and strong capability to model long-range dependencies [56], [57], [58], [59]. [56] first introduced a Transformer-based architecture for CD, termed bi-temporal image Transformer (BIT), built upon a Siamese framework. BIT modeled long-range spatial–temporal dependencies by projecting bi-temporal features into a compact token space, where a Transformer encoder learns global contextual relationships and a decoder refines pixel-level features. This design effectively captured semantic change cues while maintaining high efficiency, outperforming convolution-based baselines with substantially fewer parameters. Bandara and Patel [59] proposed Change-

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Former, a fully Transformer-based Siamese architecture that eliminates convolutional encoders to directly capture long-range bi-temporal dependencies. By combining a hierarchical Transformer encoder with a lightweight multi-layer perceptron (MLP) decoder, it effectively extracted multiscale contextual features, achieving superior accuracy over previous SOTA CD methods.

Nevertheless, these supervised approaches inherently rely on large-scale, high-quality annotated datasets, which are costly and labor-intensive to collect. In CD, the annotation challenge is even greater due to the requirement of pixel-level labeling for temporally aligned image pairs under diverse seasonal or illumination conditions. To address these limitations, recent studies have shifted toward SSL approaches that leverage large-scale unlabeled data.

B. SSL for CD

Early studies on SSL for CD explored the use of pseudo-labels indicating potential change regions [31], [32], [33], [34], [35], [36], [37], [38]. In these approaches, initial change maps are generated using conventional techniques such as image differencing, clustering, or pretrained segmentation models. These pseudo-labels are then used to provide the model with more explicit signals about where changes are likely to occur. Geng et al. [31] proposed an early pseudo-labeling approach in which initial change maps are generated through image differencing followed by fuzzy c-means (FCM) clustering. These maps are treated as credible samples, and pixel-wise change probabilities are derived to guide the training of the CD network. Similarly, Zong et al. [35] applied FCM to different maps and introduced a wavelet-transform-based Siamese network to enhance robustness against noise while preserving discriminative information for binary CD. To improve pseudo-label reliability, Zhao et al. [37] presented a weakly supervised framework that leverages image-level annotations. Specifically, activation maps were extracted from bi-temporal images using a feature extractor trained with weak labels and used to generate refined pseudo-change labels. This strategy effectively bridges the performance gap between fully and weakly supervised CD settings, achieving competitive results. While these methods can extract change-related features in the absence of labeled data, their performance heavily depends on the quality of the initial pseudo-labels. Noisy or inaccurate pseudo-labels can bias the model toward incorrect patterns, ultimately compromising semantic consistency in the learned representations.

Following the remarkable success of CL in the computer vision domain, several studies have introduced CL-based frameworks into the CD domain [18], [19], [20], [21], [22], [23], [24], [25]. In these approaches, the model is trained to maximize the agreement between feature representations of positive pairs while pushing apart those of negative pairs, thereby learning a discriminative embedding space without manual annotations. Several studies [18], [19] adopt CL by applying multiview augmentations to bi-temporal image pairs, encouraging the encoder to learn representations that are invariant to appearance variations. In particular, Chen and Bruzzone [19] introduced an alignment mechanism to enhance superpixel-level consistency across views, enabling more structured feature learning in the absence of dense annotations. To incorporate local semantic variations into the contrastive framework, Chen et al. [20] proposed a synthetic fusion strategy, where building-instance masks were extracted from auxiliary labels and composited onto the input image, thereby creating object-aware augmented views. Jiang et al. [23] further extend the framework to global–local CL by explicitly modeling both coarse-level semantics and pixel-level features. In a related but orthogonal direction, graph-based CL has been explored for CD by constructing superpixel-induced graphs and contrastively optimizing structural relationships among graph vertices [26]. Such approaches aim to suppress pixel-level noise and exploit local structural consistency, particularly for heterogeneous remote sensing images, by performing CL within graph representations rather than relying solely on pixel-wise features.

While several recent studies have introduced multiview-based CL frameworks into CD by adopting bi-temporal inputs, these approaches remain fundamentally oriented toward learning view-invariant representations rather than explicitly modeling change-related features. In CD tasks, bi-temporal images are typically spatially aligned, and the variations of interest correspond to localized structural changes, whereas variations unrelated to actual changes are mainly induced by seasonal effects, illumination differences, atmospheric conditions, or radiometric inconsistencies. In contrast, conventional multiview augmentations in CL intentionally introduce artificial geometric perturbations and spatial misalignment, which are treated as irrelevant variations during representation learning. This mismatch limits the ability of standard multiview CL pipelines to capture fine-grained change cues. To compensate for this limitation, existing multiview augmentation-based studies [18], [19], [20] incorporate additional constraints, such as explicit use of bi-temporal image pairs, superpixel-level alignment, or auxiliary supervision (e.g., building masks or temporal correspondence cues), to construct meaningful positive and negative samples. In addition to the limitations of multiview augmentation, representations learned under conventional CL objectives are typically optimized at a coarse semantic level, often resulting in spatial inconsistencies when independently extracted temporal features are applied to pixel-level CD tasks. Consequently, the determination of change is typically deferred to downstream components. In these stages, change information relies on explicit feature comparison or fusion through a decoder, highlighting a broader limitation of contrastive learning in directly capturing change-sensitive representations.

With the emergence of Transformer architectures and their strong capability for modeling global dependencies, mask-based generative methods such as MIM have been introduced into the CD domain [27], [28], [29], [30]. Zhang et al. [27] proposed a multimodal self-supervised pretraining framework inspired by masked autoencoders [15], using RGB–elevation data under a Vision Transformer backbone. A plug-in temporal fusion module based on masked cross-attention was further introduced to transfer pretrained representations effectively to downstream CD tasks. Wang et al. [28] also adopted an

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MIM strategy to train an encoder that reconstructs masked patches from bi-temporal inputs. Notably, they proposed a cross-temporal feature interaction module to capture temporal dynamics and contextual dependencies across acquisition times, thereby enhancing the consistency of learned change representations. Wan et al. [30] presented SSL CD, which leverages the intrinsic structure of multitemporal data to disentangle invariant representations through an intrinsic structure learning strategy. In addition, a cross-reconstruction mechanism extended MIM to better capture temporal dependencies, promoting contextual consistency in feature space.

The synergy between MIM and Transformer backbones is particularly notable, as the self-attention mechanism facilitates modeling both local and global context while preserving fine-grained details. Such dense representations are beneficial for pixel-level prediction tasks. However, since these methods reconstruct each image independently, they do not explicitly encode change-related features during pretraining and thus may lack sufficient discriminative power. As a result, change localization is deferred to either a Siamese feature comparison stage or a task-specific decoder designed to extract change information. Recently, generative approaches have been combined with pseudo-labeling strategies to directly guide the change-related representations in a self-supervised manner [39], [40]. Zhou et al. [39] proposed STMNet, which treats masked patches—originally designed for reconstruction in MIM—as synthetic change regions. A multiscale mask change simulation strategy was introduced to generate pseudo-change labels, aiming to guide the model toward learning dense and change-related representations. Similarly, Lv et al. [40] presented S3FCD, a method that synthesizes pseudo-changes by replacing local regions of the input image with patches sampled from a remote sensing memory bank. This enables the generation of diverse change patterns, allowing the model to learn dense representations under various synthetic scenarios. These approaches share a conceptual similarity with the proposed DiPaSSCD framework in that they both aim to provide dense and change-oriented supervision signals in a self-supervised manner.

However, two key research gaps remain to be addressed. First, the simulated changes often exhibit artificial boundaries due to patch-wise masking or replacement, which differ significantly from the spatially coherent patterns observed in real-world land-cover transitions. Such artificial transitions may create overly simplified learning targets, making it easier for the model to overfit to trivial differences. Second, these methods lack mechanisms to enforce global contextual discrimination. Since the training objective is typically limited to local reconstruction loss, the models are less capable of capturing broader appearance variations—such as seasonal or atmospheric changes—that often lead to false positives (FPs) in CD.

III. Methodology

Fig. 2 illustrates the overall framework of the proposed DiPaSSCD, which consists of three key components: 1) OBL augmentation; 2) DiPaCL; and 3) downstream CD.

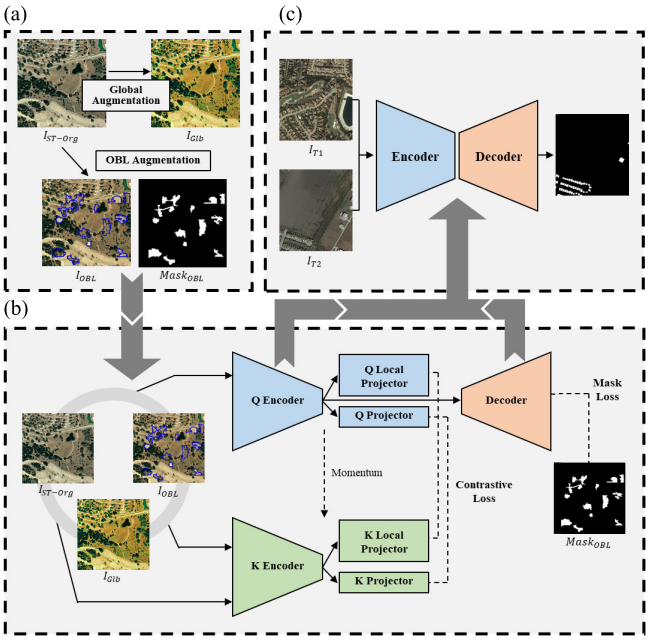


Fig. 2. Overall framework of the proposed DiPaSSCD. It comprises three major components. (a) OBL augmentation for generating difference-paired samples. (b) DiPaCL-based SSL pretraining that learns change-related representations. (c) Downstream fine-tuning for CD.

A. OBL Augmentation

Most existing CL approaches primarily rely on generating two randomly augmented views from a single data sample to construct positive pairs, while negative pairs are drawn from augmentations of different images. Although multiview augmentation enforces invariance against appearance variations, its underlying assumptions conflict with the task of CD. CD focuses on identifying subtle and localized differences between spatially aligned bi-temporal images, whereas conventional CL relies on global photometric distortions and spatially inconsistent crops that inevitably introduce artificial misalignment. As a result, such augmentation strategies can suppress change-related signals, ultimately hindering the learning of discriminative representations for local change regions.

To address this limitation, we propose an OBL augmentation strategy based on OBIA and local augmentation. The method introduces localized transformations by randomly sampling and applying segment-level pasting from a single temporal image, thereby generating perturbations that mimic local variations relevant to CD. The overall OBL augmentation procedure is summarized in Algorithm 1 and Fig. 3. Formally, the inputs consist of a single temporal RGB image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, to which a 20-pixel buffer is applied around the boundary to avoid incomplete segments during sampling, together with the following parameters: the number of segments N, the number of selections n, the compactness c, the min-size factor gamma. For the input image, OBIA segmentation is applied to extract homogeneous land-cover regions, using the simple linear iterative clustering (SLIC) superpixel algorithm [60]. SLIC is a widely used superpixel method that adapts k-means

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Algorithm 1 OBL Augmentation

INPUT:
RGB image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, number of segments N, number of selections n, compactness c, min-size factor γ
OUTPUT:
Augmented image $\mathbf{I}' \in \mathbb{R}^{H\times W\times 3}$, binary mask $\mathbf{M} \in \{0,1\}^{H\times W}$

$\mathbf{I}' \leftarrow \mathbf{I},\ \mathbf{M} \leftarrow \mathbf{0}$
Reflect padding: $\tilde{\mathbf{I}} \leftarrow \text{REFLECTPAD}(\mathbf{I}, p = 20)$
Segmentation: $\tilde{\mathbf{S}} \leftarrow \text{SLIC}(\tilde{\mathbf{I}}, N, c, \gamma)$
$\mathbf{S} \leftarrow \tilde{\mathbf{S}}[p{:}p+H,\ p{:}p+W]$
$\mathcal{U} \leftarrow \text{UNIQUE}(\mathbf{S})$
$\mathcal{S}_{\text{sel}} \leftarrow \text{RANDOMSAMPLE}(\mathcal{U}, \min(n, |\mathcal{U}|))$
for $s \in \mathcal{S}_{\text{sel}}$ **do**
  $\mathbf{m} \leftarrow \mathbb{I}(\mathbf{S} = s)$
  $\Omega \leftarrow \{(x,y) \mid \mathbf{m}(y,x) = 1\}$
  $[x_{\min}, x_{\max}], [y_{\min}, y_{\max}] \leftarrow \text{BBOX}(\mathbf{m}_s)$
  $\mathbf{R} \leftarrow \mathbf{I}[y_{\min} : y_{\max},\ x_{\min} : x_{\max}]$
  $h, w \leftarrow \text{SHAPE}(\mathbf{R})$
  $x' \leftarrow \text{RANDOMSAMPLE}(1, W - w)$
  $y' \leftarrow \text{RANDOMSAMPLE}(1, H - h)$
  $\mathbf{T} \leftarrow \mathbf{I}'[y' : y' + h,\ x' : x' + w]$
  $\mathbf{a} \leftarrow \mathbf{m}[y_{\min} : y_{\max},\ x_{\min} : x_{\max}]$
  $\boldsymbol{\alpha} \leftarrow \text{GAUSSIANBLUR}(\mathbf{a}, \sigma)$
  $\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} / \max(\boldsymbol{\alpha})$
  $\mathbf{I}'[y' : y' + h,\ x' : x' + w] \leftarrow \boldsymbol{\alpha} \odot \mathbf{R} + (1 - \boldsymbol{\alpha}) \odot \mathbf{T}$
  $\mathbf{M}[y' : y' + h,\ x' : x' + w] \leftarrow \max(\mathbf{M}[y' : y' + h,\ x' : x' + w], \mathbf{a})$
end for
return $\mathbf{I}', \mathbf{M}$

clustering to the joint color–spatial domain. By iteratively grouping pixels based on both spectral similarity and spatial proximity, it produces compact and homogeneous segments that encode the semantics of natural image structures. Specifically, the hyperparameters of SLIC in our implementation were set as follows. For an input patch of 1024×1024 pixels, $N = 300$, $c = 5$, and $\gamma = 0.3$. The detailed analysis and justification of the SLIC hyperparameter selection for OBL augmentation are discussed in Section V-A.

From the extracted segments, a subset $\mathcal{S}\text{selected} \subset \mathcal{U}$ of n superpixels is randomly sampled from the set of unique superpixel labels $\mathcal{U}$. For each selected label $s \in \mathcal{S}\text{selected}$, the corresponding region $R_s \subset \mathbf{I}$ is extracted and pasted at a random location (x', y') in a copy of the original image. Since superpixel segments are irregularly shaped, we first compute a bounding box that encloses each segment based on the range of its x and y coordinates. The bounding box is used to extract a local region R_s while preserving the original RGB values of the pixels within the segment. Within this bounding box, the pixels belonging to the segment retain their original RGB values, whereas all other pixels are set to zero to enable the isolated composition of each local segment.

The paste location (x', y') is uniformly sampled from the feasible spatial range such that the entire region R_s of size $h_s \times w_s$ lies fully inside the image domain. When multiple pasted regions overlap, the corresponding areas are sequentially overwritten through the iterative blending process. To ensure smooth transitions between the pasted region and the surrounding context, we apply a Gaussian-blurred binary alpha mask $\alpha_s \in [0,1]^{h_s \times w_s}$, which also exposes the model to locally ambiguous boundary samples. The blended image is then obtained as

$$I'_{y':y'+h_s,\,x':x'+w_s} = \alpha_s \cdot R_s + (1 - \alpha_s) \cdot I_{y':y'+h_s,\,x':x'+w_s} \tag{1}$$

where h_s and w_s denote the height and width of the region R_s, respectively. Simultaneously, to facilitate dense-level self-supervision, a binary mask $M \in 0,1^{H\times W}$ is constructed to encode the fused regions

$$M_{y':y'+h_s,,x':x'+w_s} = M_{y':y'+h_s,,x':x'+w_s} \vee \mathbf{1}(R_s) \tag{2}$$

where $\mathbf{1}(R_s)$ denotes the binary mask of the selected superpixel and $\vee$ represents the pixel-wise logical OR operation. This binary mask enables the model to explicitly associate localized perturbations with their corresponding spatial regions during training, thereby strengthening pixel-level feature learning.

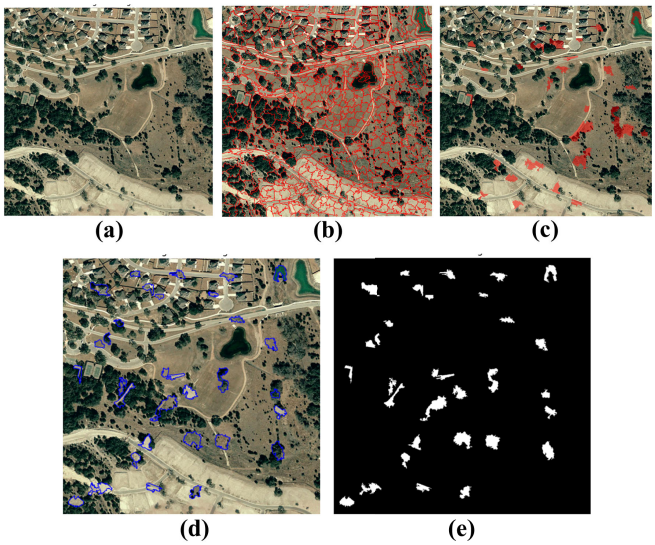


Fig. 3. Illustration of the OBL augmentation process and its results. (a) Original single-temporal image. (b) Superpixel segmentation obtained using the SLIC algorithm. (c) Randomly selected n segments. (d) Augmented OBL image with randomly fused segments (blue overlay). (e) Its corresponding binary OBL mask.

B. Difference-Paired Contrastive Learning

To exploit difference-oriented SSL for CD, we propose the DiPaCL framework (Fig. 4). Specifically, three types of input images are prepared: (1) the original input $\mathbf{I}_{\text{org}}$, (2) a globally augmented image $\mathbf{I}_{\text{glb}}$ applied color jittering, and (3) a locally augmented image $\mathbf{I}_{\text{OBL}}$ generated by the OBL augmentation. Through the combination of these three types, three synthetic bi-temporal pairs are constructed. The $(\mathbf{I}_{\text{org}}, \mathbf{I}_{\text{OBL}})$ pair, which contains local differences within the pair, is adopted as the query set. The $(\mathbf{I}_{\text{glb}}, \mathbf{I}_{\text{OBL}})$ pair, which similarly incorporates localized variations, serves as the positive key set, whereas the $(\mathbf{I}_{\text{org}}, \mathbf{I}_{\text{glb}})$ pair is employed as the negative key set.

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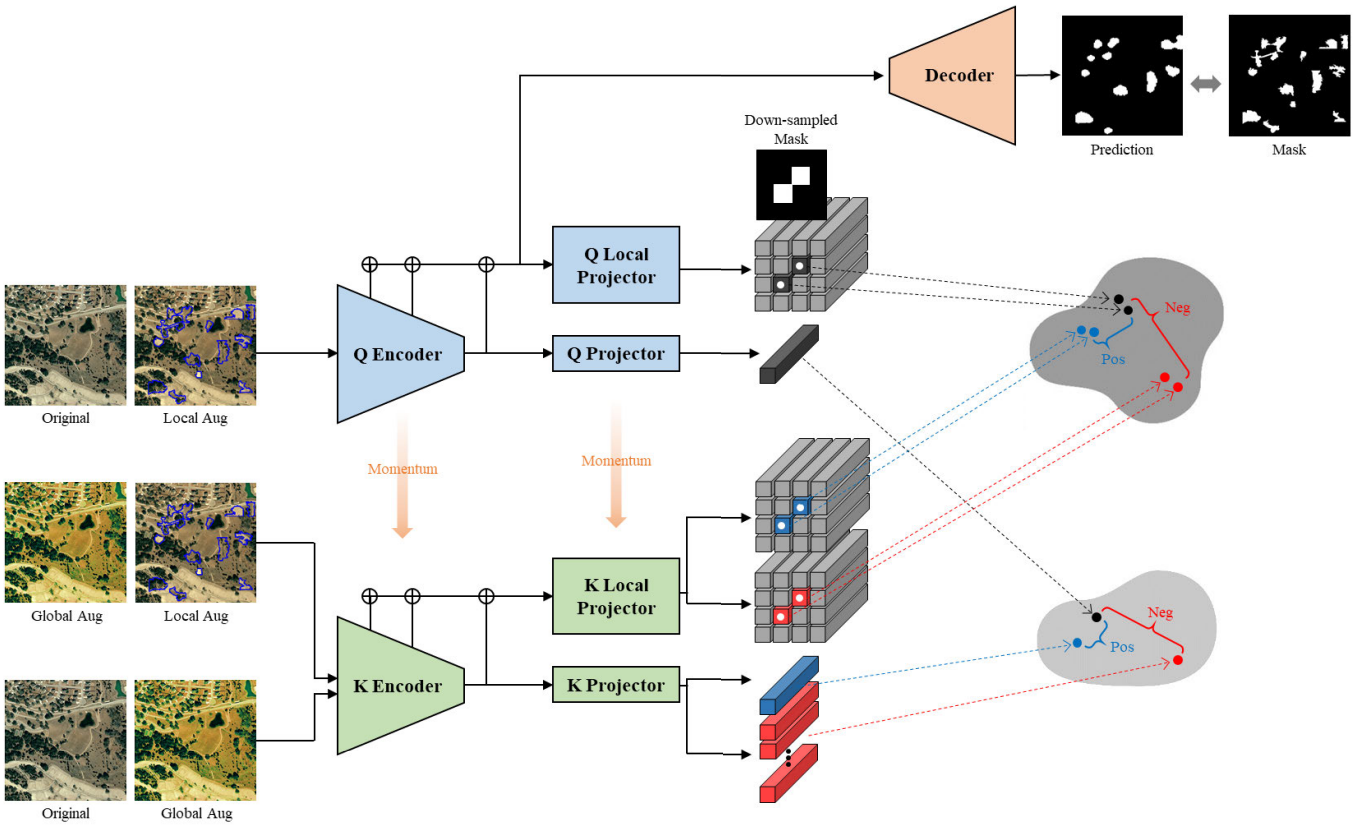


Fig. 4. Workflow of the proposed DiPaCL. Three synthetic image pairs are input into the query and key encoders. The encoder–projector outputs are embedded into both global and local feature spaces for CL. The query encoder further feeds its output into the decoder, which predicts the OBL mask for mask-guided dense supervision.

In this formulation, difference-paired inputs are constructed such that positive pairs include localized structural perturbations introduced by OBL augmentation, whereas negative pairs involve only global appearance variations. This design isolates structural differences from appearance-level changes by perturbing spatially localized regions while preserving the global scene context.

Each pair is independently encoded into query and key representations through a query encoder and a momentum-updated key encoder, respectively. Similar to MoCo v2, both query and key representations are projected into a contrastive embedding space via a projection head, which decouples the feature space for downstream CD tasks from the embedding space optimized for contrastive loss. The query encoder and its projector are updated by backpropagation, while the key encoder and projector are updated through momentum averaging from the query branch, ensuring consistent feature evolution. In the context of bi-temporal inputs with diverse global and local augmentations, the momentum encoder provides a slowly evolving reference that stabilizes representation learning across training iterations.

The negative key representations are stored in a dynamic memory bank called a queue. This memory queue maintains a diverse set of negative samples corresponding to change-irrelevant appearance variations from previous mini-batches, enabling efficient and consistent negative sampling beyond the current batch. Specifically, at each training iteration, newly computed negative key representations from the momentum encoder are enqueued, while the oldest entries are dequeued to keep the queue size fixed. During contrastive loss computation, the current query representations are contrasted against all key representations stored in the queue, which serve as negative samples. This queue-based negative sampling strategy, commonly adopted in momentum CL frameworks, allows the use of a large and diverse set of negatives without increasing the mini-batch size while ensuring representation consistency through the momentum-updated encoder. Such a design is particularly important under bi-temporal inputs, where global and local augmentations introduce substantial appearance variations across training iterations. CL alone does not rely on explicit feature differencing operations such as subtraction or concatenation. Instead, difference-related information is encoded through contrastive supervision imposed on difference-paired inputs. By enforcing representations of pairs containing localized perturbations to be pulled together while pushing apart those exhibiting only global variations, the contrastive objective biases the embedding space toward difference-sensitive features.

To strengthen dense-level representation learning in CL, we additionally introduce a mask-guided local projector that explicitly exploits the OBL masks. Specifically, multilevel feature maps extracted from the encoder are element-wise

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multiplied with the OBL mask to isolate representations within the synthetically perturbed regions. Local CL is then performed only on these masked regions. For each masked region, the representation from the $(\mathbf{I}_{\text{glb}}, \mathbf{I}_{\text{OBL}})$ pair is treated as a local positive key, as it reflects the injected local perturbation, whereas the corresponding representation from the $(\mathbf{I}_{\text{org}}, \mathbf{I}_{\text{glb}})$ pair serves as a local negative key, since no local variation is present in that region. For computational efficiency, the local contrastive objective uses batch-level negatives without maintaining a separate memory queue.

These localized samples are then employed to learn similarity embeddings that guide the model to focus on regions with synthetic perturbations. This design complements the global contrastive objective by combining coarse-level CL loss at the image scale with local CL loss at the patch level, as detailed in the following. In this way, the model is encouraged to map representations of pairs with localized differences closer together while pushing apart those that share only global variations.

While CL is effective in providing strong discriminative power, it is less effective in explicitly encoding spatially localized change information, which is critical for CD tasks involving small or subtle variations. To address this limitation, we integrate CL with a mask-guided dense prediction strategy that directly employs change-related supervision into the feature learning process. Specifically, we leverage the OBL masks as pseudo-labels to provide explicit supervision on the locations and extents of synthetic changes. By adopting an encoder–decoder architecture, only the encoder features corresponding to the $(\mathbf{I}_{\text{org}}, \mathbf{I}_{\text{OBL}})$ pair—containing synthetic local variations—are forwarded to the decoder. The decoder learns to predict pixel-level change responses, which are directly compared against the OBL mask. Through this process, change information can be encoded at the dense feature level, as the model is forced to discriminate perturbed regions from unchanged ones under pixel-level supervision. This explicit supervision complements the contrastive objective, which biases the embedding space toward difference-sensitive representations at a global and local level. As a result, DiPaCL jointly encodes difference information through both contrastive pairing and mask-guided dense prediction. The corresponding loss functions are detailed in the following.

For the encoder backbone, we adopt the feature exchange mechanism of the Changer architecture [54], which was originally designed to enhance cross-temporal feature interaction. In particular, its ChangerEx module facilitates fine-grained alignment between bi-temporal features, which is beneficial for recognizing subtle local variations introduced by OBL.

Unlike conventional SSL approaches, where the encoder learns invariant representations from single images and the independently represented features are compared within the Siamese architecture, SOTA CD models increasingly incorporate complex bi-temporal interactions within the encoder to capture subtle differences. Such architectural complexity makes it difficult to directly adopt previous SSL schemes. In contrast, DiPaCL is inherently difference-oriented, as it takes paired images with explicit differences as input and learns embeddings aligned with the objectives of CD models. This alignment enables DiPaCL to serve as a general pretraining paradigm for a wide range of existing and future CD architectures.

C. Loss Function

The proposed DiPaCL framework is optimized by jointly leveraging a contrastive learning loss and an OBL mask-guided loss. This design ensures that both global semantic consistency and local change discrimination are effectively captured during training.

First, the global contrastive loss (L_{GC}) is formulated following the InfoNCE objective adopted in MoCo:

$$\mathcal{L}_{\text{GC}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\frac{q_{g,i}\cdot k_{g,i}^{+}}{\tau} - \log\left(\exp\left(q_{g,i}\cdot k_{g,i}^{+}/\tau\right) + \sum_{k^{-}\in\mathcal{Q}_g}\exp\left(q_{g,i}\cdot k^{-}/\tau\right)\right)\right] \tag{3}$$

where $q_{g,i}$ denotes the ith global query, $k_{g,i}$ is the corresponding positive key, Q_g is the negative sample queue, N is the batch size, and T is the temperature parameter. Both $q_{g,i}$ and $k_{g,i}$ are extracted from difference-paired inputs matching $\mathbf{I}_{\text{OBL}}$ with $\mathbf{I}_{\text{org}}$ or $\mathbf{I}_{\text{glb}}$, which explicitly contain local ifferences. Consequently, the network learns to push these pairs closer in the feature space while simultaneously increasing their distance from negative samples that do not contain local differences.

Similar to the global contrastive loss, a local contrastive loss (L_{LC}) is also defined using InfoNCE

$$\mathcal{L}_{\text{LC}} = -\frac{1}{|\tilde{\mathcal{M}}|}\sum_{i\in\tilde{\mathcal{M}}}\left[\frac{q_{l,i}\cdot k_{l,i}^{+}}{\tau} - \log\left(\exp\left(q_{l,i}\cdot k_{l,i}^{+}/\tau\right) + \sum_{k^{-}\in\mathcal{Q}_l}\exp\left(q_{l,i}\cdot k^{-}/\tau\right)\right)\right] \tag{4}$$

where $q_{l,i}$ and $k_{l,i}$ represent the ith local query and positive key, respectively; Q_l denotes the set of negative keys within the batch; and $\tilde{M}$ represents the set of pixel indices in which the downsampled OBL mask takes the value of 1, corresponding to the regions identified as changes. Positive pairs are defined based on features aligned with mask-indicated changed regions, while negative pairs correspond to unchanged regions. Unlike the global branch, the local contrastive loss does not utilize a memory queue for efficiency reasons, and the comparisons are restricted to features within the mini-batch.

Both contrastive objectives are backpropagated through the query encoder and projector, while the key encoder remains detached from gradient updates. Following the momentum update mechanism of MoCo, the key encoder parameters are updated as an exponential moving average of the query encoder, thereby maintaining consistency in the dictionary representations.

Also, a mask-guided loss (L_M) is introduced to exploit the change-related signals from OBL-generated masks. Specifically, the prediction from the difference-paired query set

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$(\mathbf{I}_{\text{org}}, \mathbf{I}_{\text{OBL}})$ is compared against the ground-truth OBL mask via cross entropy

$$\mathcal{L}_M = -\frac{1}{H \times W} \sum_{u=1}^{H} \sum_{v=1}^{W} \left[M^{(u,v)} \log \hat{Y}^{(u,v)} + \left(1 - M^{(u,v)}\right) \log \left(1 - \hat{Y}^{(u,v)}\right) \right] \tag{5}$$

where $M \in \{0,1\}^{H \times W}$ denotes the binary OBL mask and $\hat{Y}$ is the prediction. The gradient of $\mathcal{L}_M$ is propagated to the query encoder and decoder, while the projector and key encoder are detached from this optimization.

Finally, the overall loss function is formulated as a weighted combination of the three objectives

$$\mathcal{L}_{\text{total}} = \lambda_{\text{glb}} \mathcal{L}_{\text{GC}} + \lambda_{\text{loc}} \mathcal{L}_{\text{LC}} + \lambda_{\text{mask}} \mathcal{L}_M \tag{6}$$

where λ_{glb}, λ_{loc}, and λ_{mask} are hyperparameters that balance the contributions of global CL, local CL, and mask-guided dense prediction, respectively. This unified optimization strategy enables DiPaCL to capture both fine-grained and change-related representations.

D. Downstream CD

Through DiPaCL pretraining, both the encoder and decoder are equipped with the ability to represent dense features that are sensitive to change-related patterns. However, since the OBL augmentation introduces diverse types of changes, the self-supervised model alone cannot guarantee precise discrimination of task-specific target changes. Therefore, the pretrained DiPaCL model is required to be additionally fine-tuned on domain-specific CD datasets via transfer learning.

Given a bi-temporal image pair $(X_{\text{pre}}, X_{\text{post}})$, the pretrained encoder–decoder network produces a dense change prediction map

$$\hat{Y} = f_{\text{dec}} \left(f^{q}_{\text{enc}} \left(X_{\text{pre}}, X_{\text{post}} \right) \right) \tag{7}$$

where f denotes the DiPaCL model parameters transferred to the downstream task. In contrast to conventional Siamese-based approaches, which independently encode each temporal image and subsequently compare their features, DiPaCL directly learns from difference-oriented representations. Such a design not only enhances the ability to capture subtle and localized variations but also improves compatibility with diverse and sophisticated CD architectures.

For transfer learning, the objective function is defined as the cross entropy between the predicted change map and the target label

$$\mathcal{L}_{\text{CD}} = -\sum_{i} Y^{(i)}_{\text{target}} \log \hat{Y}^{(i)} \tag{8}$$

where Y_{target} denotes the ground-truth change mask provided by each dataset. In this way, the pretrained DiPaCL model is aligned with domain-specific objectives and utilized for downstream CD.

IV. Experiments and Results

A. Datasets

In this study, the proposed DiPaSSCD framework was validated on three widely used open-source CD datasets, LEVIR-CD [61], S2Looking [62], and DSIFN [63]. With the aim of verifying the effectiveness of the self-supervised pretraining, we further introduced a data-limited setting by restricting the available training samples to 10% of the full dataset.

LEVIR-CD [61] is a large-scale building CD dataset constructed from Google Earth VHR imagery. It consists of 637 bi-temporal image pairs with a size of 1024 × 1024 pixels and a spatial resolution of approximately 0.5 m. The images were acquired over multiple urban regions across Texas between 2002 and 2018, covering diverse urbanization patterns. The dataset provides high-quality binary change annotations focusing on building footprints, making it a standard benchmark for urban CD. Following the official split, the dataset is divided into 445 pairs for training, 64 pairs for validation, and 128 pairs for testing. In the data-limited scenario, we restricted the training set by randomly sampling 45 pairs among the original 445 pairs, which were fixed for all experiments, while the validation and testing sets were identical to the standard split.

S2Looking [62] is a recently released large-scale dataset specifically designed for a building CD from side-looking satellite imagery. Unlike conventional benchmarks that predominantly rely on near-nadir observations, S2Looking provides more diverse perspectives by incorporating off-nadir acquisitions. The dataset comprises 5000 bi-temporal image pairs, each with a size of 1024 × 1024 pixels and a spatial resolution of 0.5–0.8 m. The images were collected between 2017 and 2020 from multiple VHR satellites, including GaoFen (GF), SuperView (SV), and BeiJing-2. The ground-truth labels offer pixel-level building-change annotations, covering both newly constructed and demolished regions. In this study, we followed the official split of 70% for training, 10% for validation, and 20% for testing. In addition, to simulate data-limited scenarios, we further constrained the training data by randomly selecting 350 pairs while keeping the validation and testing sets identical to the official partition.

DSIFN [63] is a building CD dataset manually collected from Google Earth imagery, covering six urban areas in China, namely, Beijing, Chengdu, Shenzhen, Chongqing, Wuhan, and Xi'an. Five city pairs (Beijing, Chengdu, Shenzhen, Chongqing, and Wuhan) are cropped into 394 subimage pairs with a spatial size of 512 × 512 pixels, while the remaining city pair, Xi'an, is cropped into 48 image pairs and exclusively used for testing. This study followed the official data split, in which the augmented dataset consists of 3600 image pairs for training, 340 image pairs for validation, and 48 image pairs for testing.

B. Comparison Methods

To evaluate the effectiveness of the proposed DiPaSSCD, we compared it with a diverse set of SOTA CD models and representative SSL approaches, including FC-EF [49],

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FC-Siam-conc [49], FC-Siam-diff [49], SNUNet-CD [50], TinyCD [55], ChangerEX [54], ChangeFormer [59], BiT [56], as well as SSL-based MoCo v2 [6], SimSiam [9], and BEiT [14]. Each comparative approach is briefly introduced as follows.

1) *FC-EF [49]:* This approach is a supervised CD method based on the UNet architecture. This model adopts an early fusion strategy, where bi-temporal images are concatenated along the channel dimension and fed into a fully convolutional network. Owing to its simplicity, FC-EF is commonly regarded as a baseline in DL-based supervised CD studies.
2) *FC-Siam-Conc [49]:* In this model, a Siamese encoder extracts features independently of the bi-temporal images, which are subsequently concatenated before the decoding stage. This design preserves temporal independence while enabling interaction at the decoding stage.
3) *FC-Siam-Diff [49]:* Similar to FC-Siam-Conc, this model employs a Siamese encoder but computes the absolute difference between bi-temporal feature maps. The subtraction operation highlights temporal discrepancies, often improving sensitivity to subtle changes.
4) *SNUNet-CD [50]:* Extending the UNet architecture, SNUNet-CD integrates densely connected skip pathways within a Siamese design. This facilitates rich multiscale fusion of shallow and deep features, thereby improving boundary localization and small CD in very-high-resolution imagery.
5) *TinyCD [55]:* This model introduces a lightweight Siamese UNet-based structure to reduce parameter counts while maintaining competitive accuracy. In addition, attention-based fusion modules are integrated at the early fusion stage to enhance feature correspondence between the bi-temporal images.
6) *ChangerEX [54]:* ChangerEX incorporates parameter-free feature exchange layers to facilitate cross-temporal interactions between encoder streams. When combined with alignment modules, it improves the modeling of temporal discrepancies without increasing model complexity.
7) *BiT [56]:* Inspired by the recent success of Transformers, BiT adopts Transformer-based encoders to jointly model spatial and temporal contexts in paired images. It explicitly learns cross-temporal dependencies, providing an early exploration of Transformer architectures for CD tasks.
8) *ChangeFormer [59]:* Based on a hierarchical Transformer backbone, ChangeFormer captures both local details and long-range dependencies in bi-temporal inputs. The architecture improves contextual modeling and demonstrates strong performance across multiple CD benchmarks.
9) *M-CD [64]:* M-CD is a Mamba-based CD framework that leverages state space models to efficiently capture long-range dependencies in bi-temporal remote sensing images. By replacing self-attention with Mamba blocks, the architecture achieves linear-time complexity while maintaining an expanded receptive field, enabling effective modeling of large-scale spatial contexts. This design allows M-CD to balance computational efficiency and representational capacity, leading to improved change segmentation performance across multiple benchmarks.
10) *Change3D [65]:* Change3D reformulates bi-temporal CD as a video modeling problem by treating the input image pair as a two-frame video. By introducing learnable perception frames and leveraging a lightweight video encoder, the framework enables direct temporal interaction between frames, allowing change-related information to be implicitly captured without relying on task-specific change extractors. This unified design is effective across multiple CD tasks.
11) *MoCo v2 [6]:* MoCo is a widely used SSL framework that maintains a dynamic memory queue and a momentum encoder for CL. The later version improves the original design by incorporating stronger data augmentations and an additional prediction head, which substantially stabilizes training and enhances representation quality. The original MoCo is designed to learn invariant representations from single images. MoCo serves as a representative CL approach that relies on multiview augmentation to enforce invariance to spatial perturbations. As discussed earlier, such multiview augmentation promotes robustness to spatial misalignment, which is not well aligned with the requirements of CD tasks. To verify the effectiveness of the proposed OBL augmentation, MoCo is therefore selected as a baseline for comparative evaluation. To extend this framework to CD, a pretrained MoCo v2 model is adapted by employing a double-stream architecture followed by a decoder, and the entire network is fine-tuned on CD datasets.
12) *SimSiam [9]:* SimSiam is an SSL approach based on noncontrastive learning, which eliminates the need for negative pairs by learning representations through asymmetric prediction between augmented views. Similar to MoCo, SimSiam relies on multiview augmentation to generate training pairs, encouraging invariance to view-specific perturbations. Accordingly, SimSiam is selected as an additional SSL baseline to further evaluate the effectiveness of the proposed OBL augmentation. Likewise, a pretrained SimSiam model is adapted to the CD task by employing a double-stream architecture followed by a decoder, and fine-tuning is performed on CD datasets.
13) *BEiT [14]:* BEiT employs an MIM strategy with discrete visual tokens. Specifically, images are first tokenized into visual codewords using a pretrained tokenizer, and a proportion of the tokens are masked before being reconstructed by a Vision Transformer. To adapt this self-prediction SSL model with generative capability for masked regions to the CD task, this study performs fine-tuning on CD datasets.

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TABLE I
QUANTITATIVE COMPARISON OF PRECISION, RECALL, AND $F1$-SCORE (IN %) UNDER THE 10% LABELED CONDITION ON THE LEVIR-CD, S2LOOKING, AND DSIFN DATASETS

Model	LEVIR (10%)			S2Looking (10%)			DSIFN (10%)		
	Precision	**Recall**	**F1**	**Precision**	**Recall**	**F1**	**Precision**	**Recall**	**F1**
FC-EF [49]	52.8241	24.8874	33.8343	22.9511	0.8639	1.6651	70.0357	22.6865	34.4788
FC-Siam-conc [49]	71.8506	40.9440	52.1630	53.7348	9.8520	16.6512	**76.2037**	15.1215	25.2354
FC-Siam-diff [49]	66.2808	49.3915	56.6031	**74.9076**	17.6524	28.5718	57.6937	32.5531	41.6216
TinyCDv2 [55]	83.5144	67.4930	74.6538	59.4156	40.9364	48.4746	49.8881	51.0995	50.4865
SNUNet [50]	85.2325	68.6685	76.0591	55.1581	44.6364	49.3426	53.1913	43.6823	47.9701
ChangerEx [54]	79.2359	**78.7767**	79.0057	59.7363	45.3717	51.5724	29.7076	**76.1211**	42.7365
BiT [56]	85.4688	70.6378	77.3488	64.4092	44.5404	52.6631	40.0392	59.2757	47.7945
ChangeFormer [59]	73.6289	69.6848	71.6026	59.9689	31.0443	40.9104	41.6527	42.8713	42.2532
M-CD [64]	71.7842	49.6213	58.6798	54.5873	18.3421	27.4579	35.7996	58.1502	44.3163
Change3D [65]	71.1575	63.7203	67.2339	57.7377	43.8368	49.8361	41.2444	47.1849	44.0151
MoCo [6]	84.8683	61.4520	71.2864	54.7064	44.3306	48.9750	62.8325	37.8682	47.2560
SimSiam [9]	82.2629	70.3069	75.8164	61.9821	37.7562	46.9270	58.0556	40.3319	47.5973
BEiT [14]	56.7167	28.9347	38.3200	50.6365	24.5882	33.1024	42.6602	55.1872	48.1218
Proposed	**87.4646**	75.7226	**81.1712**	61.0496	**48.8190**	**54.2536**	55.4287	46.9520	**50.8394**

C. Experimental Setting

To quantitatively evaluate the performance of the proposed framework, three widely used metrics in CD were employed: precision, recall, and $F1$-score. Precision and recall enable a comprehensive assessment of commission errors, which correspond to FP detections, and omission errors, which correspond to false negative (FN) detections. The $F1$-score further provides an integrated measure that reflects the overall detection accuracy in a balanced manner.

All experiments were conducted using PyTorch 2.1.0 with CUDA 11.8 on an NVIDIA RTX A6000 GPU equipped with 48-GB memory. The input size of the model was set to 512×512 pixels, where each sample was randomly cropped from the original 1024×1024 patches during data augmentation.

In the SSL stage, we adopted a stochastic gradient descent (SGD) optimizer with a momentum of 0.9 and a weight decay of 1×10^{-4}. The base learning rate was initialized at 0.03 and scheduled using cosine annealing with linear warm-up. Also, the batch size was set to 64, the queue size was set to $K = 8192$, the momentum update coefficient was set to $m = 0.999$, the temperature was set to $\tau = 0.2$, and the feature dimension was set to 128. The loss weights were set to 0.3, 0.3, and 0.4 for the global contrastive loss, the local contrastive loss, and the mask-guided loss, respectively. The weighting factors were empirically selected based on validation experiments, and we observed that the model performance remained stable under moderate variations of these values, indicating that the proposed framework does not rely on precise loss-weight tuning. The mask-guided loss, which directly supervises change-related information and corresponds to the primary objective of the proposed method, was therefore assigned a slightly higher weight, while the other loss terms act as auxiliary regularization components.

During fine-tuning, the batch size was adjusted according to the proportion of training data. For the full dataset, a batch size of 64 was used, whereas smaller settings with 10% and 5% labeled samples were trained with a batch size of 8. The same optimizer, learning rate, and scheduling strategy as in the SSL stage were adopted to ensure consistency across experiments. All comparative methods were trained under identical experimental conditions, including optimizer, learning rate, scheduler, and batch size, to ensure a fair comparison with the proposed framework.

D. Experimental Results

1) Results on LEVIR-CD: Table I summarizes the quantitative results of different CD models on the LEVIR-CD dataset under a limited-data setting, where only 10% of the training samples were used. This experimental scenario highlights the robustness of each method when trained with scarce supervision.

Among the supervised baselines, early double-stream CNN-based architectures such as FC-EF, FC-Siam-conc, and FC-Siam-diff achieved relatively low $F1$-scores of 33.83%, 52.16%, and 56.60%, respectively. Their recall values were considerably lower than precision, which indicates that the models suffer from substantial omission error. In contrast, other SOTA models that exploit bi-temporal feature fusion strategies demonstrated clear improvements, consistently yielding higher $F1$-scores. Specifically, Transformer-based approaches, including ChangeFormer and BiT, achieved $F1$-scores of 69.68% and 70.63%, respectively, outperforming earlier CNN-based models. In contrast, the Mamba-based M-CD achieved a lower $F1$-score of 58.68%, indicating that two-frame CD tasks may benefit more from spatial feature modeling than from temporal sequence modeling. Among all comparison models, ChangerEX achieved the highest $F1$-score of 79.01%, which can be attributed to its explicit feature interaction mechanism for capturing change-related cues.

Both contrastive and noncontrastive SSL approaches, including MoCo v2 and SimSiam, achieved $F1$-scores exceeding 70%, whereas the MIM approach BEiT yielded a substantially lower score of 38.32%. This performance gap indicates that self-prediction alone has limited discriminative capability for CD, particularly under data-limited settings, where the large parameter capacity of Transformer-based models poses challenges for effective fine-tuning. In comparison, the proposed DiPaSSCD achieved the highest $F1$-score of 81.17%, outperforming all supervised and SSL-based CD

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models evaluated in this study. Unlike MoCo and SimSiam, which primarily learn coarse-level representations from spatially misaligned multiview augmented pairs and rely on downstream fine-tuning to capture fine-grained change cues, DiPaSSCD integrates object-level masks and pairwise difference information directly into the SSL stage through the proposed OBL augmentation. This design enables the model to explicitly capture change-relevant local variations from spatially aligned bi-temporal image pairs.

As a result, the proposed framework outperformed strong supervised state-of-the-art methods, including ChangerEX, BiT, and SNUNet, by 2.17, 3.82, and 5.11 percentage points in $F1$-score, respectively, demonstrating the effectiveness of OBL in addressing the limitations of conventional multiview augmentation for CD.

Meanwhile, the proposed method exhibits a slightly lower recall compared to ChangerEX, despite achieving the highest precision and $F1$-score. This behavior can be attributed to the characteristics of the LEVIR-CD dataset, which is dominated by densely built-up urban areas where change pixels are widely distributed across the scene. Such conditions generally favor a relatively permissive decision boundary to capture numerous dense changes. Through pretraining, DiPaCL encourages a more regularized and less dataset-specific decision boundary, thereby mitigating overspecialization to densely distributed change patterns. As a result, a modest increase in omission errors may be observed for certain models. However, this tradeoff is compensated by a substantial reduction in commission errors caused by overestimation, ultimately leading to the highest overall $F1$-score among the compared methods.

2) Results on S2Looking: Table I presents the comparative results on the S2Looking dataset, which consists of off-nadir satellite image pairs acquired under side-looking viewing conditions. Owing to the severe off-nadir viewing conditions, all methods yielded generally lower performances than those obtained on LEVIR-CD.

Similar to the previous results, early double-stream CNN-based models, including FC-EF, FC-Siam-conc, and FC-Siam-diff, yielded $F1$-scores below 30%. Their recall values were significantly lower than precision, indicating severe omission errors. In particular, FC-EF achieved an $F1$-score of only 1.67%, indicating that the model failed to effectively extract meaningful change information from off-nadir imagery under limited training data conditions. Likewise, the Mamba-based M-CD also exhibited notably low performance under the 10% labeled condition, consistent with its behavior on the LEVIR-CD dataset, which further indicates the limited effectiveness of sequence-oriented modeling in data-scarce bi-temporal scenarios.

Among the more advanced supervised approaches, the Transformer-based BiT achieved the highest performance with an $F1$-score of 52.66%, followed closely by ChangerEX at 51.57%. These results confirm the advantage of feature-interaction and Transformer-based architectures in handling the challenging side-looking conditions.

Regarding SSL approaches, a similar performance pattern was observed on LEVIR-CD. The CL methods, MoCo v2 and SimSiam, achieved higher $F1$-scores than the self-prediction-based method BEiT, which exhibited limited discriminative ability under data-limited conditions. The proposed DiPaSSCD achieved the highest $F1$-score of 54.25%. This result corresponds to improvements of 1.59, 2.68, and 4.91 percentage points over BiT, ChangerEX, and SNUNet, respectively, thus demonstrating the robustness of the proposed framework in challenging off-nadir scenarios.

In more detail, the proposed method exhibits slightly lower precision than FC-Siam-diff, BiT, and SimSiam on the S2Looking dataset, in contrast to the behavior observed on LEVIR-CD. This difference can be explained by the characteristics of S2Looking, which is predominantly composed of rural regions where change labels are sparse, favoring a narrower decision boundary. In this context, DiPaCL acts as a regularizing mechanism by preventing excessive adaptation to dataset-specific characteristics during training. While this may introduce additional commission errors in certain cases, it enables the model to recover a larger number of previously missed changes. Consequently, a substantial improvement in recall is achieved, leading to the best overall $F1$-score on the S2Looking dataset.

3) Results on DSIFN: Table I also presents the comparative results on the DSIFN dataset. Unlike LEVIR-CD, which constructs training and testing sets by splitting image pairs from the same urban areas, DSIFN was constructed to evaluate generalization performance by designating an entirely different city as the test set. As a result, this dataset poses a significantly more challenging setting, where distribution shifts between training and testing data lead to unstable performance across models.

Consistent with the results observed on LEVIR-CD and S2Looking, early double-stream CNN-based models exhibit relatively low $F1$-scores on the DSIFN dataset. In particular, their recall values are substantially lower than precision, indicating severe omission errors. Across the comparison methods, however, the balance between precision and recall varies considerably; some models, such as ChangerEX and M-CD, tend to overpredict change regions, while others underpredict them. This inconsistent behavior suggests that the large domain discrepancy between training and testing cities in DSIFN leads to unstable model training, resulting in highly variable inference characteristics across different architectures. Such instability highlights the challenging nature of DSIFN, where models struggle to maintain consistent error distributions under significant domain shifts. Among the comparison methods, TinyCDv2 achieves the highest $F1$-score of 50.48%, indicating that lightweight architectures can be relatively more robust to domain discrepancies by reducing overfitting and adapting more effectively to unseen urban environments.

The proposed DiPaSSCD achieves the best overall performance with an $F1$-score of 50.83%. By leveraging robust change-related representations learned during the self-supervised pretraining stage, DiPaSSCD enables more stable fine-tuning even with limited labeled data and under significant domain discrepancies. Notably, this improvement is achieved despite adopting ChangerEX as the backbone architecture, which exhibits extremely low precision when trained in a

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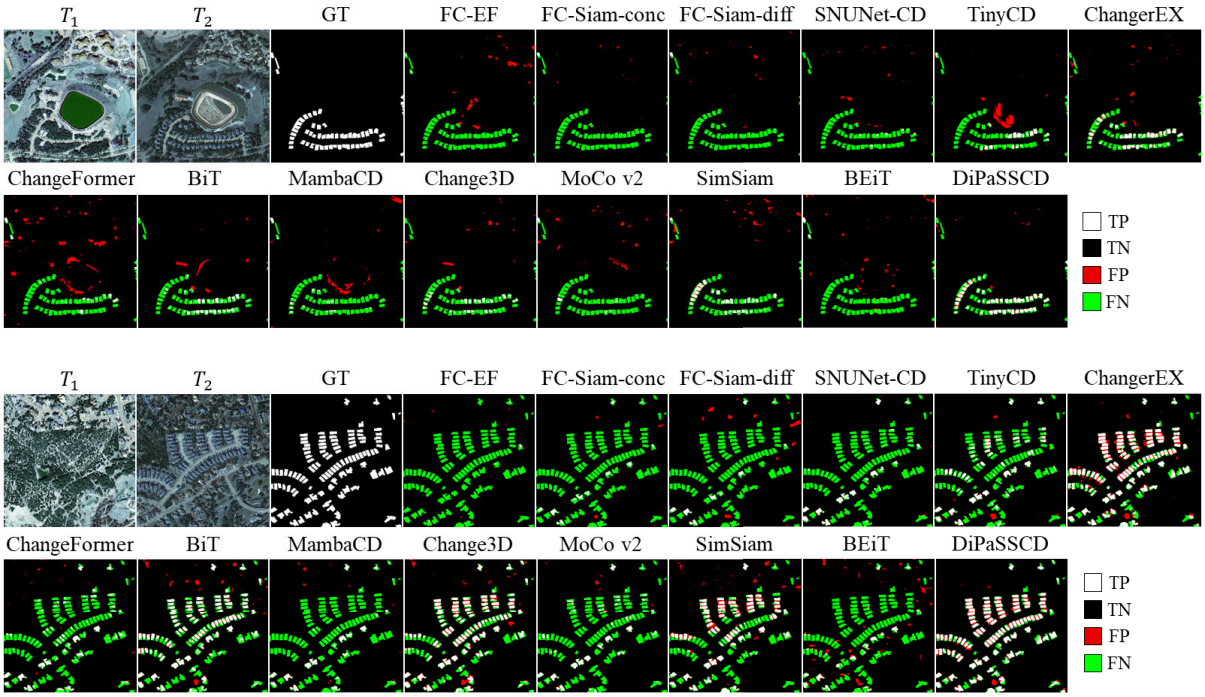


Fig. 5. Qualitative comparison of CD results on the LEVIR-CD dataset. Each example includes bi-temporal images, ground-truth maps, and predictions from 13 different models. In the visualization, white, black, red, and green denote TPs, true negatives (TNs), FPs, and FNs, respectively.

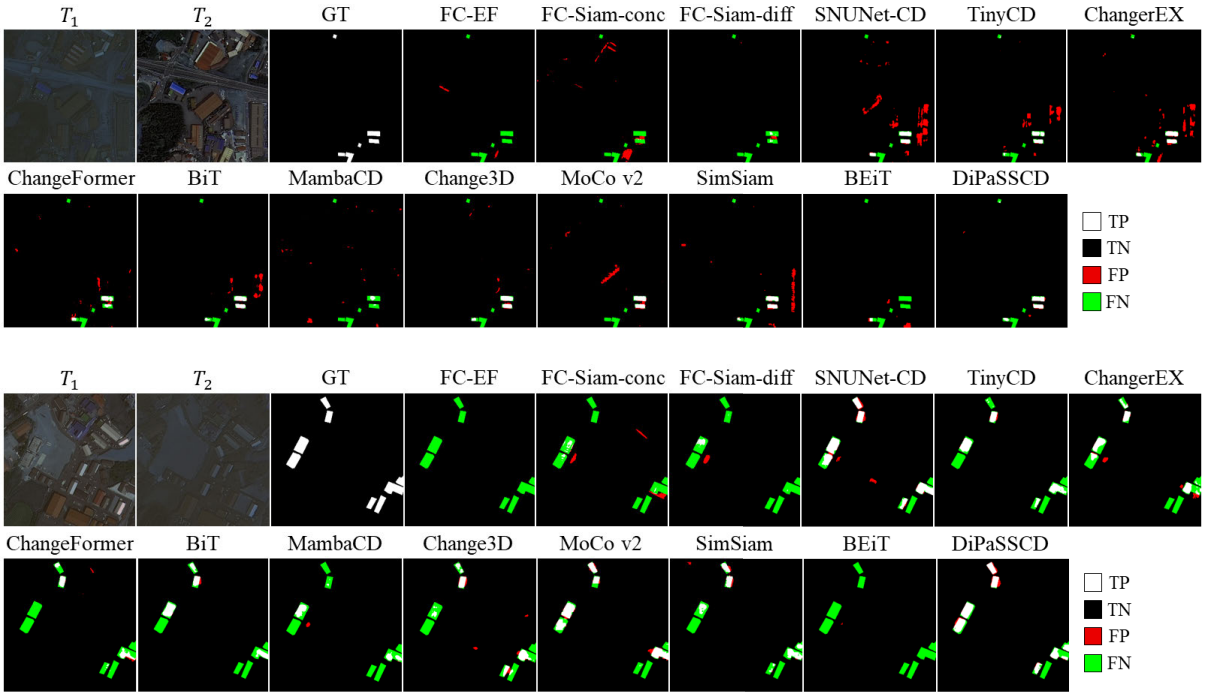


Fig. 6. Qualitative comparison of CD results on the S2Looking dataset. Each example includes bi-temporal images, ground-truth maps, and predictions from 13 different models. In the visualization, white, black, red, and green denote TPs, TNs, FPs, and FNs, respectively.

fully supervised manner. These results demonstrate that the proposed pretraining strategy effectively alleviates error imbalance and mitigates performance degradation caused by domain shifts, thereby significantly enhancing the robustness of CD in cross-city scenarios.

The regularization effects of DiPaCL observed on LEVIR-CD and S2Looking are most evident on the DSIFN dataset, which exhibits a substantial domain gap between the training and test sets and is therefore particularly prone to overfitting. Under such conditions, different models demonstrate

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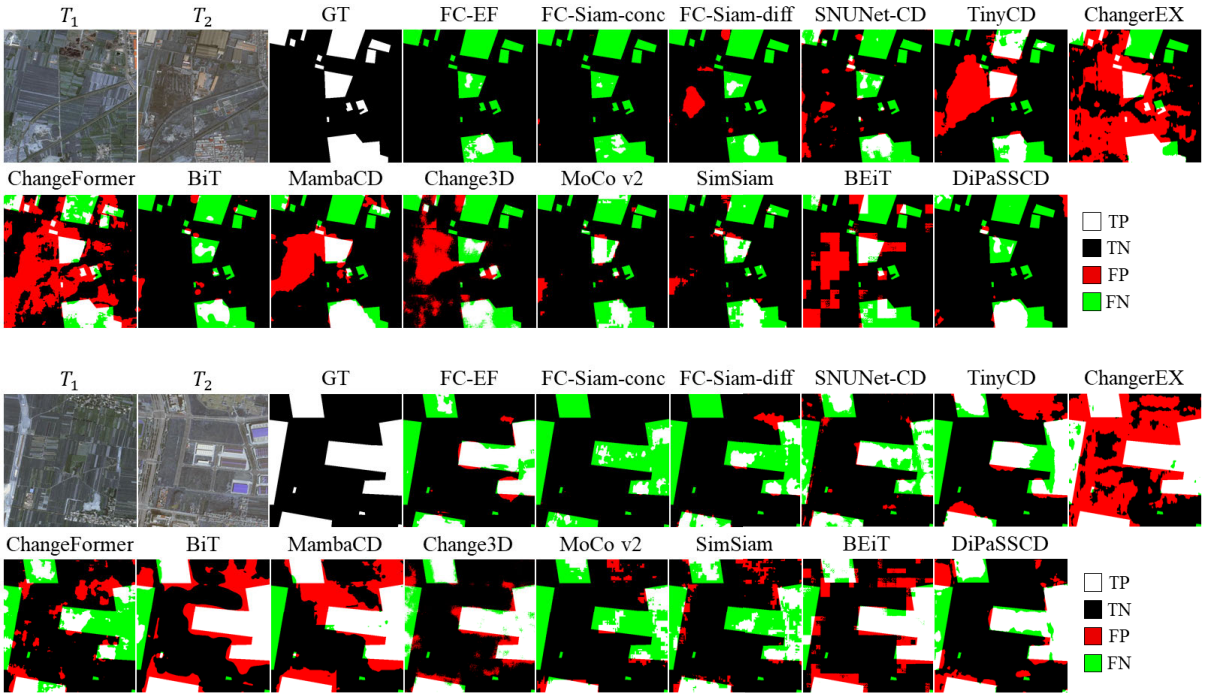


Fig. 7. Qualitative comparison of CD results on the DSIFN dataset. Each example includes bi-temporal images, ground-truth maps, and predictions from 13 different models. In the visualization, white, black, red, and green denote TPs, TNs, FPs, and FNs, respectively.

highly inconsistent precision–recall behaviors. By promoting a more stable and less biased decision boundary, DiPaCL alleviates overfitting under limited supervision and consistently improves the balance between precision and recall, ultimately achieving the highest $F1$-scores across models.

4) Qualitative Results: In addition to the metric-based quantitative evaluation, Figs. 5–7 present the qualitative comparisons of change maps generated by SOTA CD models and the proposed DiPaSSCD on the LEVIR-CD, S2Looking, and DSIFN datasets, respectively. The visual results are consistent with the quantitative findings, showing that the proposed DiPaSSCD produces more accurate predictions than existing methods under both datasets.

For the LEVIR-CD dataset, most existing approaches, including FC-EF, FC-Siam-conc, FC-Siam-diff, ChangeFormer, SNUNet, M-CD, MoCo v2, and BEiT, fail to capture building changes in densely developed areas, resulting in severe omission errors represented as FNs (in green) in Fig. 5. By contrast, the proposed DiPaSSCD correctly detects such changes as true positives (TPs), demonstrating enhanced sensitivity to fine-scale structural variations in dense urban environments, owing to its ability to learn local-scale representations.

For the S2Looking dataset, conventional CD models exhibit numerous commission errors, represented as FPs (in red) in Fig. 6. These errors are largely attributed to illumination variations across temporal images, where differences in roof reflectance are incorrectly detected as changed areas. In contrast, the proposed method significantly reduces FPs through DiPaCL-based pretraining with diverse synthetic perturbations, which allows the model to distinguish irrelevant appearance variations from real land-cover changes. Furthermore, compared to the existing models that exhibit FNs indicated by green overlays, the proposed method successfully converts these regions into TPs (white overlays), thereby successfully reducing omission errors.

For the DSIFN dataset, the qualitative results in Fig. 7 reveal that most comparison methods suffer from substantial missed detections, accompanied by pronounced commission errors. In particular, ChangerEX and MambaCD exhibit a large number of FPs, highlighted in red, indicating a tendency to overpredict change regions. Such errors can be attributed to overfitting under the data-scarce training condition, which exacerbates the impact of domain discrepancies between the training and testing cities and leads to unstable inference on the unseen test set. In contrast, the proposed method significantly suppresses FPs while maintaining sensitivity to building changes, resulting in fewer missed detections. This improvement is primarily attributed to the DiPaCL-based pretraining strategy, which exposes the model to diverse synthetic perturbations and encourages the learning of robust change-related representations. Notably, while many baseline models exhibit biased predictions toward either overestimation or underestimation of changes, the proposed framework achieves a more balanced tradeoff between FPs and FNs, leading to more reliable and visually consistent CD results on DSIFN.

V. Discussion

A. Sensitivity Analysis of SLIC Hyperparameters for OBL Augmentation

The characteristics of superpixel segments generated by SLIC, including their shape, internal homogeneity, and spatial

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extent, are highly dependent on the choice of hyperparameters. Since the proposed OBL augmentation explicitly relies on superpixel segments as basic units for synthesizing artificial changes, the validity and effectiveness of the generated changes are inherently influenced by the SLIC parameter determination. Among the SLIC hyperparameters, the number of segments N and the compactness parameter c play a dominant role in determining the segmentation behavior. The compactness parameter controls the relative importance between spectral similarity and spatial proximity during segmentation. As c increases, spatial regularization becomes more dominant, resulting in superpixels with more regular and grid-like shapes, often approaching square forms. The number of segments N determines the overall granularity of the superpixel decomposition. Larger values of N generally lead to smaller and more homogeneous segments; however, excessively fine segmentation may introduce severe fragmentation, producing noisy superpixels that lack semantic meaning. In contrast, overly coarse segmentation may cause a single superpixel to encompass multiple land-cover types, violating the object-level assumption underlying object-based augmentation strategies.

These two parameters jointly affect not only the geometry of the extracted superpixels but also the spatial patterns of the synthetic changes generated by OBL augmentation. The OBL augmentation adopts an object-based paradigm in which land-cover regions are automatically segmented and relocated to synthesize artificial changes. This approach implicitly assumes that: 1) each segmented object corresponds to a relatively homogeneous land-cover region and 2) the pasted region introduces a meaningful land-cover discrepancy with respect to its target location. If the superpixel size is excessively large or constrained to overly regular shapes, heterogeneous land-cover types may be mixed within a single segment. Also, overly small superpixels may result in noisy and fragmented synthetic changes that lack semantic significance, thereby providing weak or invalid supervision for change representation learning. Based on these considerations, this study conducts a focused analysis at the preprocessing stage to investigate the sensitivity of the proposed OBL augmentation to SLIC hyperparameters. Specifically, we examine how different configurations of N and c affect: 1) the heterogeneity of superpixel segments and 2) the ratio of invalid synthetic changes produced by OBL augmentation.

The segment heterogeneity is defined to quantify the internal spectral consistency of superpixel segments and is formulated as

$$\mathcal{H} = \frac{1}{|S|} \sum_{s \in S} \frac{\mathrm{Var}(\mathbf{I}_s)}{\mathrm{Var}(\mathbf{I})} \tag{9}$$

where S denotes the set of all superpixel segments, $\mathbf{I}_s$ represents the set of pixels belonging to segment s, and $\mathbf{I}$ denotes the entire image. This metric measures the average normalized intrasegment variance, with lower values indicating higher spectral homogeneity within superpixels. In addition, the invalid change ratio is introduced to assess the validity of synthetic changes generated by OBL augmentation and is computed as

$$\mathcal{R}_{\text{invalid}} = \frac{|\{p \in \Omega_m \mid \|\mathbf{I}(p) - \mathbf{I}'(p)\| < \sigma(\mathbf{I})\}|}{|\Omega_m|} \tag{10}$$

where $\mathbf{I}$ and $\mathbf{I}'$ denote the original and augmented images, respectively; Ω_m indicates the augmentation mask region; and $\sigma(\mathbf{I})$ is the global standard deviation of the original image. This ratio represents the proportion of pixels within the augmentation mask that exhibit negligible spectral discrepancy from the original image and is used to identify semantically invalid synthetic changes.

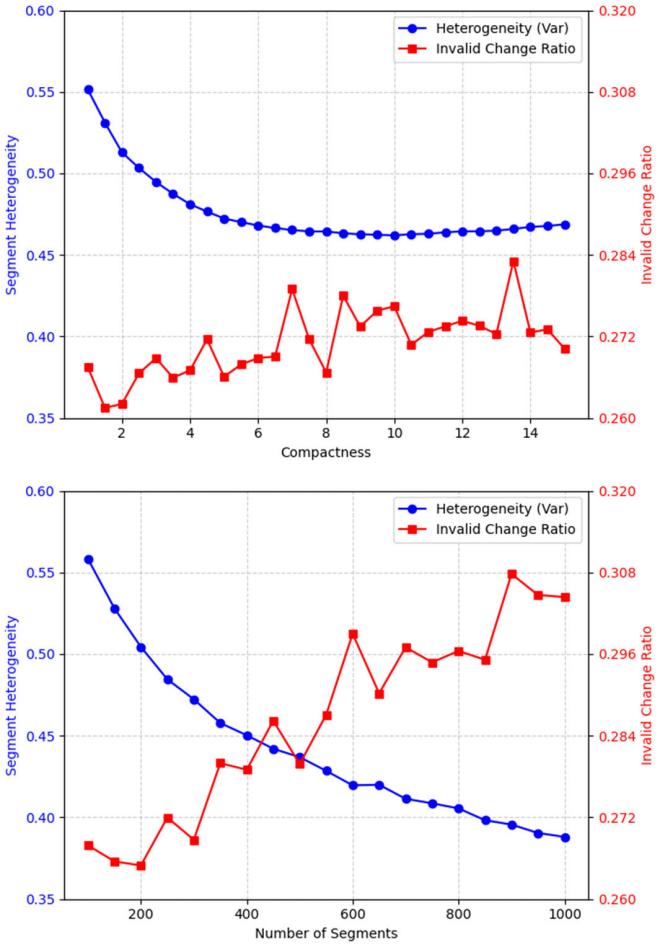


Fig. 8. Sensitivity analysis of SLIC hyperparameters for OBL augmentation. The figure illustrates the relationships between the compactness parameter c and the number of segments N with respect to superpixel heterogeneity and the invalid change ratio. These results analyze how SLIC parameter choices influence segmentation homogeneity and the validity of synthetically generated changes, providing guidance for selecting stable and balanced hyperparameters in the proposed OBL augmentation framework.

Fig. 8 illustrates the effects of SLIC hyperparameters, including c and N, on the segmentation heterogeneity and invalid change ratio metrics. As shown in Fig. 8, the segment heterogeneity exhibits increased values when c is either excessively small or large. For large values of c, spatial proximity becomes dominant over spectral similarity, forcing superpixels to adopt regular and spatially constrained shapes. This behavior increases the likelihood that spectrally distinct land-cover types are merged into a single segment, leading

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to higher intrasegment heterogeneity. Conversely, when c is too small, spatial constraints are weakened, allowing spatially distant regions with similar spectral characteristics to be merged. This effect results in the formation of abnormally large segments that encompass multiple land-cover types, again increasing heterogeneity. Notably, the increase in heterogeneity becomes more pronounced when c falls below 4. While heterogeneity values remain relatively stable within the range of approximately 0.45–0.47 for $c \geq 4$, they rise sharply to values exceeding 0.55 for smaller c. The invalid change ratio associated with compactness exhibits a relatively irregular trend but shows a slight increasing tendency as c increases. Larger values of c produce more spatially constrained and fragmented superpixels, which in turn tend to generate synthetic changes with limited semantic validity. Although a weak positive correlation between c and the invalid change ratio is observed, the absolute variation remains modest, ranging from approximately 0.26 to 0.29. Overall, both heterogeneity and the invalid change ratio remain stable with respect to the choice of the compactness parameter when $c \geq 4$. Considering the gradual increase in invalid change ratio with larger c, this study selects $c = 5$ as a balanced choice that maintains segmentation stability while avoiding excessive fragmentation.

Fig. 8 presents the relationship between the number of segments N and segment heterogeneity. Heterogeneity decreases as N increases since finer superpixel granularity reduces the likelihood of multiple land-cover types being grouped within a single segment. Specifically, heterogeneity values decrease approximately linearly from 0.55 to 0.40 as N increases. In contrast, the invalid change ratio shows a gradual increasing trend with increasing N. Excessively fine segmentation results in highly fragmented superpixels, which tend to generate noisy and semantically uninformative synthetic changes during OBL augmentation. This behavior is consistent with the trend observed for large compactness values. Overall, the number of segments N introduces a clear tradeoff between segmentation homogeneity and synthetic change validity. While increasing N improves spectral consistency within superpixels, it simultaneously increases the proportion of invalid synthetic changes. Nevertheless, the overall variation of both metrics remains within a moderate range (below 0.5 for heterogeneity and below 0.3 for invalid change ratio), indicating robust parameter sensitivity. In this study, N is set to 500 to balance segmentation quality and augmentation validity.

B. Adaptability of DiPaCL Across Diverse CD Architectures

The proposed DiPaCL allows for flexible adaptation to a wide range of CD models since its difference-paired setting and change-related feature representation are inherently consistent with the operational mechanism of supervised bi-temporal CD models. Unlike conventional contrastive frameworks that are restricted to single-image encoders and subsequently adapted through Siamese-based fine-tuning, the proposed approach is more agnostic to architectural constraints. In particular, recent CD backbones increasingly incorporate intensive bi-temporal feature interactions within the encoder stage, while the design of DiPaCL provides a more compatible initialization for diverse CD architectures.

TABLE II

PERFORMANCE COMPARISON OF FC-SIAM DIFF [49], CHANGEREX [54], BIT [56], AND CHANGEFORMER [59] WITH AND WITHOUT DIPACL UNDER 10% AND ALL DATA SETTINGS ON LEVIR-CD, S2LOOKING, AND DSIFN DATASETS

Dataset	Model	DiPa	10%			All		
			Precision	Recall	F1	Precision	Recall	F1
LEVIR	FCSD	wo	66.2808	49.3915	56.6031	90.7069	87.3625	89.0033
		w	72.5892	51.5176	60.2646	92.1629	86.4437	89.2118
		Δ	-	-	+3.6615	-	-	+0.2085
	Changer EX	wo	79.2359	78.7767	79.0057	93.3948	89.2623	91.2818
		w	87.4646	75.7226	81.1712	92.6845	90.4195	91.5380
		Δ	-	-	+2.1655	-	-	+0.2562
	Change Former	wo	73.6289	69.6848	71.6026	92.2176	89.4282	90.8015
		w	79.2162	66.4150	72.2530	91.8249	89.2102	90.4987
		Δ	-	-	+0.6504	-	-	-0.3028
	BiT	wo	85.4688	70.6378	77.3488	92.4142	89.2951	90.8278
		w	84.9324	72.5458	78.2519	93.0731	89.2619	91.1277
		Δ	-	-	+0.9031	-	-	+0.2999
S2 Looking	FCSD	wo	74.9076	17.6524	28.5718	73.8975	23.7947	35.9981
		w	76.5298	20.3530	32.1546	81.3868	26.9693	40.5136
		Δ	-	-	+3.5828	-	-	+4.5155
	Changer EX	wo	59.7363	45.3717	51.5724	69.7335	54.6790	61.2955
		w	61.0496	48.8190	54.2536	68.4325	58.2465	62.9300
		Δ	-	-	+2.6812	-	-	+1.6345
	Change Former	wo	59.9689	31.0443	40.9104	69.7392	54.3339	61.0821
		w	68.0782	41.9201	51.8889	69.2853	55.4975	61.6297
		Δ	-	-	+10.9785	-	-	+0.5476
	BiT	wo	64.4092	44.5404	52.6631	70.6875	54.7010	61.6751
		w	67.8087	45.1397	54.1994	70.7023	57.2476	63.2676
		Δ	-	-	+1.5363	-	-	+1.5925
DSIFN	FCSD	wo	57.6937	32.5531	41.6216	69.5822	47.2043	56.2493
		w	61.6908	38.2760	47.2413	71.1225	50.3159	58.9367
		Δ	-	-	+5.6197	-	-	+2.6874
	Changer EX	wo	29.7076	76.1211	42.7365	63.3060	58.2055	60.6487
		w	55.4287	46.9520	50.8394	67.2993	55.2487	60.6815
		Δ	-	-	+8.1029	-	-	+0.0328
	Change Former	wo	41.6527	42.8713	42.2532	61.3647	55.7754	58.4367
		w	36.9782	56.9863	44.8241	68.1995	54.4795	60.5723
		Δ	-	-	+2.5709	-	-	+2.1356
	BiT	wo	40.0392	59.2757	47.7945	53.1332	68.1297	59.7041
		w	46.5887	49.8939	48.1847	52.0319	65.4052	57.9571
		Δ	-	-	+0.3902	-	-	-1.7470

To validate this property, we conducted experiments on four representative models: 1) FC-Siam-diff [49], a classical Siamese-based CNN; 2) ChangerEx [54], a CNN-based architecture with explicit feature interaction; and 3) two Transformer-based approaches, BiT [56] and ChangeFormer [59]. Table II quantitatively summarizes the results on the LEVIR-CD, S2Looking, and DSIFN datasets under both limited (10%) and full (100%) training label conditions.

The results demonstrate that DiPaCL yields consistent performance gains across all models when only 10% of annotated training pairs are available. The improvement is particularly pronounced on the more challenging S2Looking and DSIFN datasets, where baseline accuracies are relatively low. For instance, on the S2Looking dataset under the 10% training condition, ChangeFormer achieves an improvement of 10.98 percentage points in $F1$-score (40.91%→ 51.89%). Similarly, on the DSIFN dataset, ChangerEX exhibits an increase of 8.10 percentage points in $F1$-score. CNN-based backbones, such as FC-Siam-diff and ChangerEX, show relatively stable performance improvements across datasets, whereas Transformer-based models, including ChangeFormer and BiT, exhibit larger performance fluctuations depending on the dataset. In contrast, when the full training set is used, the impact of DiPaCL becomes less pronounced, as the models can already exploit abundant supervised signals. A few exceptions are observed. ChangeFormer on LEVIR-CD and BiT on DSIFN experience slight performance degradations (−0.30 and −1.74 percentage points in $F1$-score, respectively). Nevertheless, all other cases exhibit nondecreasing or positive

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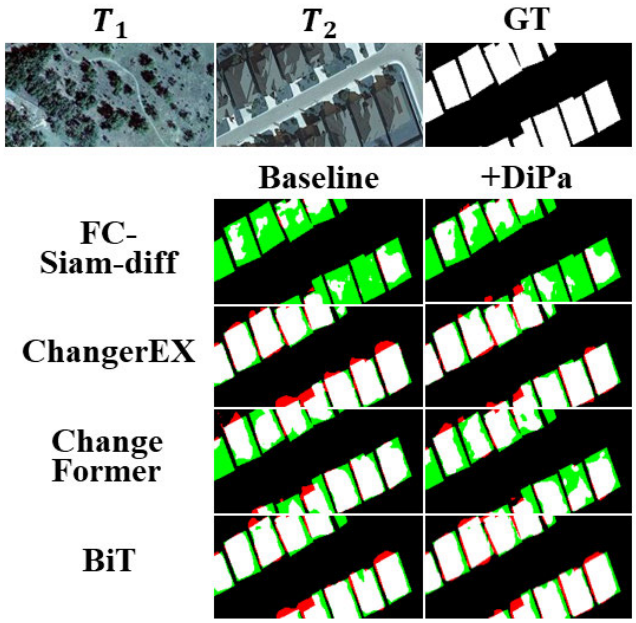


Fig. 9. Qualitative comparison of four representative CD models—FC-Siam-Diff [49], ChangerEx [54], BiT [56], and ChangeFormer [59]—on the LEVIR-CD dataset. Each shows bi-temporal images, ground-truth, baseline predictions, and predictions after applying the proposed DiPaCL pretraining. The white, black, red, and green overlays represent TP, TN, FP, and FN, respectively.

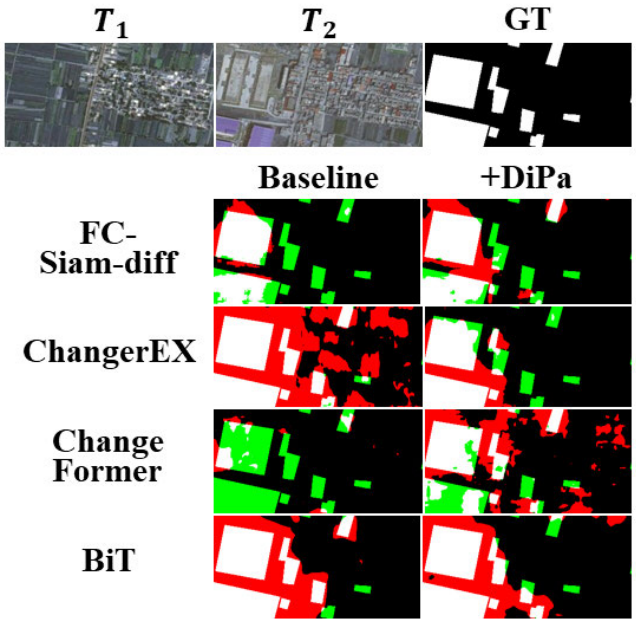


Fig. 11. Qualitative comparison of four representative CD models—FC-Siam-Diff [49], ChangerEx [54], BiT [56], and ChangeFormer [59]—on the DSIFN dataset. Each shows bi-temporal images, ground-truth, baseline predictions, and predictions after applying the proposed DiPaCL pretraining. The white, black, red, and green overlays represent TP, TN, FP, and FN, respectively.

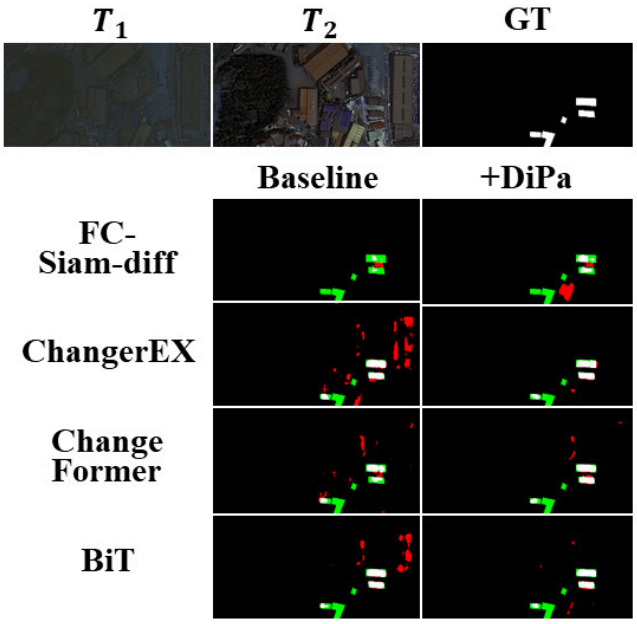


Fig. 10. Qualitative comparison of four representative CD models—FC-Siam-Diff [49], ChangerEx [54], BiT [56], and ChangeFormer [59]—on the S2Looking dataset. Each shows bi-temporal images, ground-truth, baseline predictions, and predictions after applying the proposed DiPaCL pretraining. The white, black, red, and green overlays represent TP, TN, FP, and FN, respectively.

gains, indicating that DiPaCL pretraining generally provides robust and effective initialization across diverse model architectures.

Figs. 9–11 provide qualitative evidence of the effectiveness of DiPaCL on LEVIR-CD, S2Looking, and DSIFN, respectively. The baseline models exhibit both omission and commission errors. Specifically, some building changes are missed, and the detected boundaries appear blurred, indicating limited precision. Furthermore, apparent variations induced by illumination or acquisition differences often lead to spurious detections in unchanged areas, and conventional baselines, including ChangerEx and Transformer-based models, tend to misinterpret such pseudo-changes as true building changes. In contrast, DiPaCL-pretrained models exhibit a significantly improved ability to suppress these FPs. This demonstrates that the proposed SSL effectively enhances discrimination between appearance-induced variations and genuine land-cover changes, such as newly constructed or demolished buildings.

Furthermore, DiPaCL effectively reduces FNs for FC-Siam-diff, which often fails to detect most building changes. By reinforcing sensitivity to potential change candidates, the pretrained model additionally detects true changes that were previously overlooked. Overall, these results confirm that DiPaCL serves as a versatile pretraining paradigm, effective across both CNN- and Transformer-based CD models.

C. Impact of Training Data Proportion on DiPaCL

The proposed DiPaCL-based pretraining encodes change-related priors from difference-paired image sets in advance, enabling the downstream CD models to learn more efficiently under label-limited conditions. To verify this property, we evaluated the performance improvements of DiPaCL across different proportions of annotated training data. Specifically, three label ratios were considered for fine-tuning: 5%, 10%, and 100% of the official training splits in LEVIR-CD and

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TABLE III
PERFORMANCE COMPARISON WITH AND WITHOUT DIPA ACROSS DIFFERENT LABELED DATA RATIOS ON LEVIR-CD AND S2LOOKING DATASETS

Dataset	Ratio	DiPa	Precision	Recall	F1
LEVIR	5%	w/o	74.5618	50.0043	59.8624
		w/	78.1924	51.9823	62.4487
		Δ	-	-	+2.5863
	10%	w/o	79.2359	78.7767	79.0057
		w/	87.4646	75.7226	81.1712
		Δ	-	-	+2.1655
	100%	w/o	93.3948	89.2623	91.2818
		w/	91.5380	92.6845	91.5380
		Δ	-	-	+0.2562
S2Looking	5%	w/o	64.8170	34.4682	45.0042
		w/	58.6203	44.1780	50.3847
		Δ	-	-	+5.3805
	10%	w/o	59.7363	45.3717	51.5724
		w/	61.0496	48.8190	54.2536
		Δ	-	-	+2.6812
	100%	w/o	69.7335	54.6790	61.2955
		w/	68.4325	58.2465	62.9300
		Δ	-	-	+1.6345

S2Looking. Table III summarizes the corresponding $F1$-score gains over the baselines.

Overall, DiPaCL-pretrained models consistently outperform the baselines across all data proportions. The improvement is most pronounced under limited supervision, where insufficient annotations prevent the baselines from fully exploiting the temporal context. For instance, on LEVIR-CD, DiPaCL achieves $F1$ gains of +2.59, +2.17, and +0.26 percentage points under the 5%, 10%, and 100% conditions, respectively. Similarly, on the more challenging S2Looking dataset, the improvements reach +5.38, +2.68, and +1.63 percentage points. These results clearly indicate that DiPaCL provides stronger benefits in low-label scenarios, while the effect diminishes as abundant supervised signals are available.

Qualitative comparisons in Figs. 12 and 13 further highlight this pattern. Each example displays the pre-change and post-change images, the ground-truth annotations, and the predictions of the baseline and DiPaCL-pretrained models under different label ratios. The last column presents the comparison results, where cyan overlays highlight regions in which false predictions of the baseline were corrected by DiPaCL, whereas orange overlays denote degraded cases. At the 5% setting, substantial improvements are observed, with significant reductions in both FNs, such as missed building changes, and FPs, such as incorrectly detected pseudo-changes. Notably, several changed objects that were completely missed by the baseline are additionally captured by DiPaCL. With 10% of the training data, DiPaCL also provides clear benefits, yielding more accurate building delineation and reducing commission errors. By contrast, with the full training set, the improvements are limited to subtle refinements along object boundaries. In some cases, performance even slightly degrades, including additional false alarms, reflecting that the contribution of SSL diminishes when supervised learning dominates. These findings confirm that DiPaCL is particularly advantageous when annotation resources are scarce, a setting highly relevant to real-world remote sensing applications where exhaustive pixel-level labeling is often infeasible.

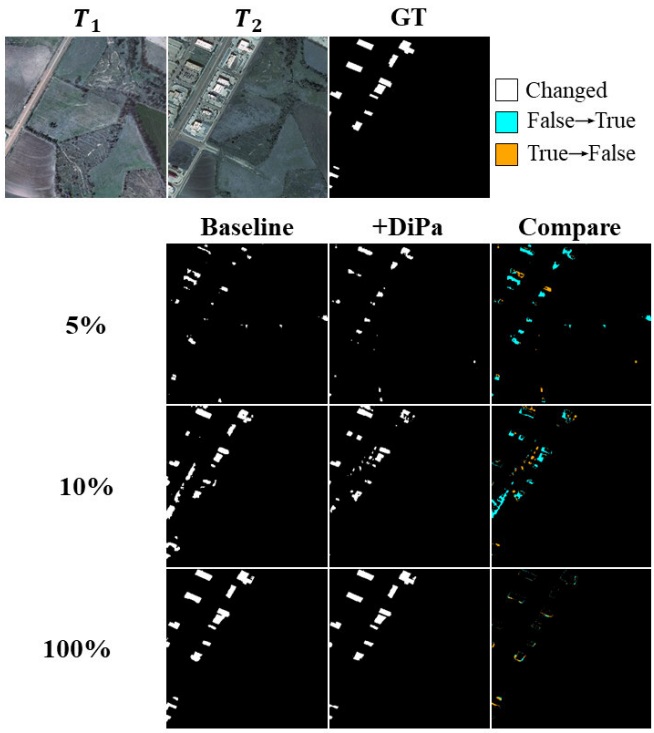


Fig. 12. Comparison of CD results under different labeled data ratios on the LEVIR-CD dataset. Each shows bi-temporal images, ground-truth, and the predictions of baseline models and their DiPaCL-pretrained counterparts. In the comparison overlay, cyan indicates improved predictions (false → true), whereas orange indicates degraded predictions (true → false) after applying DiPaCL.

D. Change-Related Feature Visualization and Analysis

Fig. 14 illustrates the feature representations learned through the proposed DiPaCL-based SSL. The first and second columns show the bi-temporal input images, followed by the corresponding ground-truth change labels in the third column. The last column presents the feature activation maps extracted from the DiPaCL-pretrained model when the bi-temporal images are input. These representations are obtained without incorporating any ground-truth change labels during SSL stages, demonstrating that DiPaCL can capture change-sensitive features in a purely self-supervised manner.

Overall, the feature maps emphasize regions where genuine changes are indicated in the ground truth, consistently assigning higher activation values to building changes than to nonchange areas. Remarkably, this behavior emerges without any supervision from ground-truth change labels during pretraining. Although the OBL augmentation relocates segments within a single image, it preserves the global radiometric distribution and visual coherence of the scene, producing synthetic images that remain natural in appearance rather than exhibiting anomalous local artifacts. Instead, the OBL and difference-paired framework of DiPaCL guides the model to encode discriminative patterns that naturally separate local change from nonchange regions.

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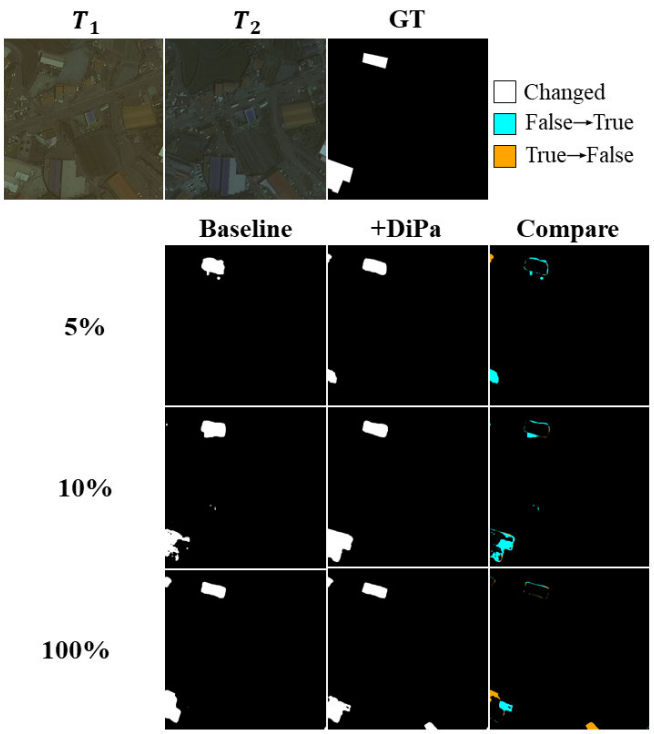


Fig. 13. Comparison of CD results under different labeled data ratios on the S2Looking dataset. Each shows bi-temporal images, ground-truth, and the predictions of baseline models and their DiPaCL-pretrained counterparts. In the comparison overlay, cyan indicates improved predictions (false → true), whereas orange indicates degraded predictions (true → false) after applying DiPaCL.

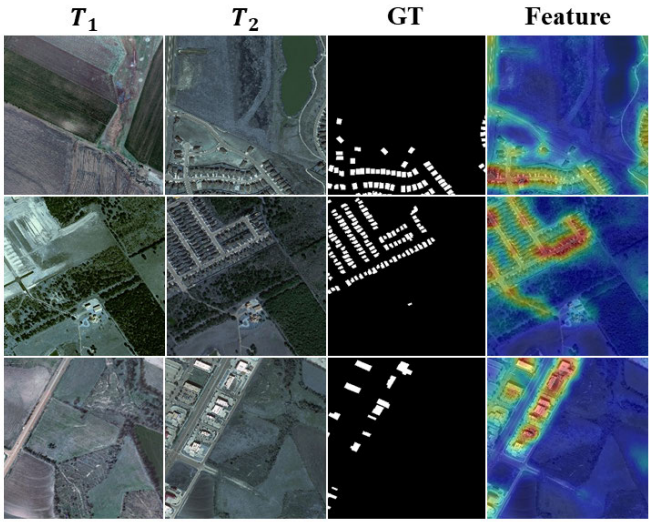


Fig. 14. Visualization of activation maps from the DiPaCL-pretrained model. Each example presents bi-temporal images, ground-truth, and corresponding feature activation maps. Warmer colors (red tones) indicate regions with higher activation, highlighting areas the model considers most relevant to change patterns.

Detailed examination indicates that building changes on previously homogeneous surfaces, such as vegetated areas, are strongly highlighted in red, whereas changes in regions with more complex spatial textures, such as cropland or heterogeneous surfaces, appear with lower activations in yellow.

TABLE IV
ABLATION STUDY ON LOSS FUNCTION CONFIGURATIONS UNDER 10% LABELED DATA ON LEVIR-CD AND S2LOOKING DATASETS

Dataset	Loss Function	Precision	Recall	F1
LEVIR_10%	L_{GC}	85.9865	73.1823	79.0694
	L_{GC}+L_{LC}	84.3605	**76.6083**	80.2977
	L_{GC}+L_{LC}+L_M	**87.4646**	75.7226	**81.1712**
S2_10%	L_{GC}	63.7642	42.2399	50.8168
	L_{GC}+L_{LC}	**64.0343**	42.3807	51.0044
	L_{GC}+L_{LC}+L_M	61.0496	**48.8190**	**54.2536**

Moreover, the model also responds to other meaningful land-cover changes not explicitly annotated in the ground truth, such as road construction, suggesting that the learned features encode diverse forms of local structural variation rather than being driven by artificial displacement cues. Since the pretraining objective is not tied to a specific semantic change category, the learned representations are inherently generic and encompass multiple change types, and the subsequent fine-tuning process is required to specialize the model for the target change classes.

Importantly, the pretrained features also demonstrate robustness against irrelevant variations. For example, apparent changes caused by illumination differences are consistently assigned low activation values, in contrast to genuine land-cover changes. This property explains why DiPaCL-pretrained models are able to suppress diverse FPs observed in the baselines, while retaining sensitivity to true structural changes. In this context, the role of SSL pretraining is not to determine the semantic validity of changes but to encourage sensitivity to local variations while suppressing irrelevant appearance fluctuations, with task-specific semantic discrimination achieved during fine-tuning under limited supervision. These observations confirm that DiPaCL effectively learns change-related representations aligned with the intrinsic objective of bi-temporal CD, enabling the model to highlight meaningful changes while suppressing pseudo-changes even without explicit supervision.

E. Loss Function Ablation Study

In addition to the conventional InfoNCE-based global contrastive loss (L_{GC}), the proposed framework incorporates two complementary loss terms: a local contrastive loss (L_{LC}) and a mask-guided loss (L_M). These components are designed to encourage the model to learn fine-grained and change-related feature representations by complementing the original CL objective.

Table IV summarizes the results of the ablation study, where three loss configurations were compared on both LEVIR-CD and S2Looking: 1) the baseline using only (L_{GC}); 2) the addition of the (L_{LC}); and 3) the proposed that further integrates (L_M). Across both datasets, the incremental inclusion of these loss functions leads to consistent $F1$-score improvements, validating the complementary role of each component. In particular, on the more challenging S2Looking dataset, the addition of (L_M) yields a significant gain of +3.25 percentage points in $F1$-score. It demonstrates that combining CL and dense prediction objectives can mitigate

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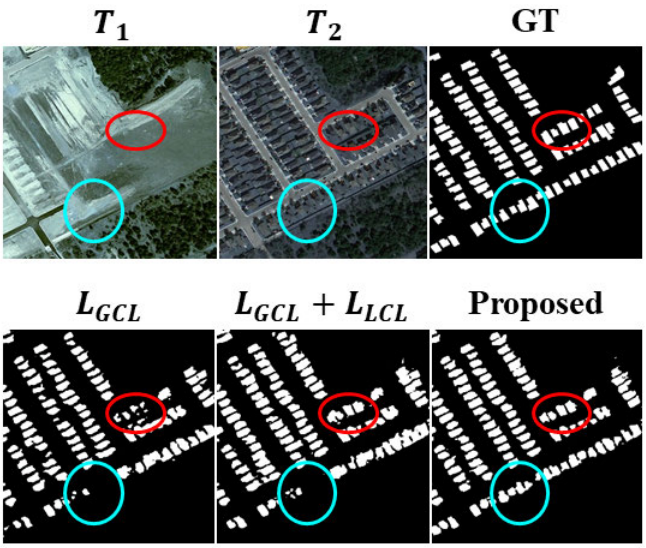


Fig. 15. Qualitative results of loss function ablation on the LEVIR-CD dataset. Each case includes bi-temporal inputs, ground-truth, and the predictions obtained with three loss configurations.

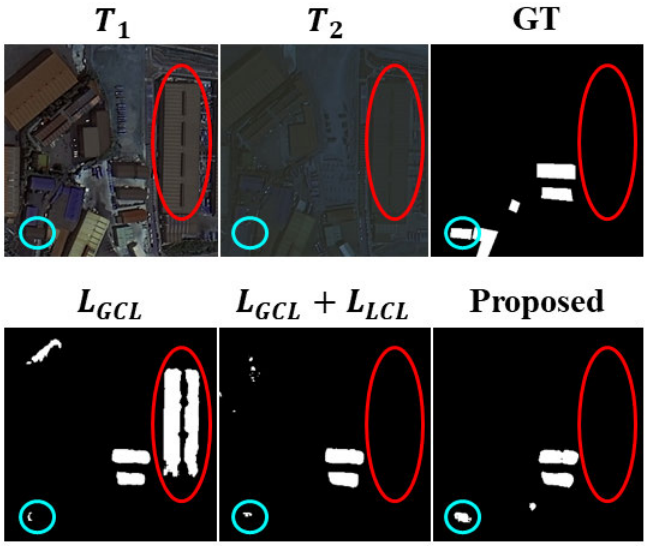


Fig. 16. Qualitative results of loss function ablation on the S2Looking dataset. Each case includes bi-temporal inputs, ground-truth, and the predictions obtained with three loss configurations.

each other's limitations. While CL enforces global discriminative representations, the mask-based dense prediction provides pixel-level change-related information, yielding more robust representations.

The qualitative comparisons in Figs. 15 and 16 further support these findings on the two datasets. As additional loss terms are incorporated, FPs are progressively suppressed, with pseudo-changes more effectively distinguished from genuine changes. Simultaneously, missed changes are increasingly recovered, indicating enhanced sensitivity to subtle structural variations. On LEVIR-CD, where buildings are densely distributed, the OBL mask-based local losses, including (L_{LC}) and (L_M), enable finer discrimination, allowing the model to separate adjacent buildings that would otherwise be merged in baseline predictions. On S2Looking, radiometric variations in roof reflectance often lead baseline models to incorrectly identify false changes. However, with the integration of the (L_{LC}), such pseudo-changes are substantially reduced, confirming that localized feature alignment helps suppress noise induced by acquisition conditions.

Overall, these results highlight the importance of jointly leveraging global discrimination, local consistency, and mask-guided prediction in the DiPaCL framework. The synergy among the three loss functions provides not only quantitative improvements but also qualitatively more reliable and interpretable change maps, thereby strengthening the robustness of the proposed pretraining strategy.

VI. Conclusion

In this study, we proposed DiPaSSCD, a difference-oriented self-supervised pretraining framework for CD from remote sensing imagery. By introducing OBL augmentation and DiPaCL, the proposed DiPaSSCD framework effectively addresses the three core limitations of existing SSL approaches: spatial misalignment from multiview augmentation, the tradeoff between discriminative and dense representations, and the lack of explicit change-related semantics.

Comprehensive experiments on the LEVIR-CD and S2Looking datasets validated the effectiveness and adaptability of DiPaSSCD. The method consistently improved $F1$-scores across CNN- and Transformer-based architectures, with the most significant benefits observed in data-limited scenarios. Furthermore, DiPaCL-based pretraining demonstrated enhanced robustness against appearance-induced pseudo-changes and improved sensitivity to subtle land-cover changes, as confirmed by both quantitative metrics and qualitative visualization. Its ability to deliver robust initialization across diverse backbones and challenging imaging conditions highlights its potential for real-world applications where annotated data are scarce.

Despite the promising results, several limitations remain. First, since DiPa is self-supervised by randomly simulating synthetic changes, it cannot explicitly differentiate among various land-cover change types, thereby requiring an additional transfer learning stage to adapt to target-specific changes. Second, the global color jittering used in the augmentation process does not fully capture the complex appearance variations observed in real-world remote sensing imagery, leading to a noticeable gap between single-temporal simulated pairs and real bi-temporal images. Third, the current evaluation has been limited to building-change datasets, and further validation across diverse land-cover types is necessary to generalize the applicability of the proposed framework. In future work, we will focus on incorporating image-translation-based radiometric augmentation and extending the evaluation to multiclass and multidomain change scenarios to further enhance the generalization and transferability of DiPa-based representations.

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ACKNOWLEDGMENT

The Institute of Engineering Research at Seoul National University, Seoul, South Korea, provided research facilities for this article.

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